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Three-dimensional Magnetotelluric Inversion based on a Data Space variant of Akaike's Bayesian Information Criterion

RUNNING HEAD:

3-D MT inversion based on DABIC

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ABSTRACT

We develop a three-dimensional (3-D) magnetotelluric (MT) inversion method based on Akaike's Bayesian Information Criterion (ABIC), which statistically determines the optimal regularization parameter that balances the prior constraints and data fitting, and therefore controls the smoothness of the final model by maximizing marginal likelihood in Bayesian statistics. To mitigate the high computational cost of calculating the ABIC indicator, we introduce a low-rank transformation, similar to that used in model-to-data space inversions, resulting in a data-space variant of ABIC. This adaptation significantly reduces computational complexity, facilitating the practical application of this statistical approach in 3-D MT inversion. Our discussion on the methodology and its application begins with a theoretical analysis of the proposed inversion method, including the underlying assumptions, the mathematical foundations of this statistical framework, a detailed derivation of the data-space ABIC formulation, and the procedure for obtaining the solution that maximizes the marginal likelihood. We then assess the method's performance using synthetic data, evaluate its stability under different inversion configurations, and benchmark it against other inversion strategies, including those based on the L-curve criterion, the cooling approach, and the original ABIC in model space. We demonstrate that the proposed inversion based on the data-space ABIC formulation yields stable performance across various configurations and offers improved objectivity compared to L-curve and cooling-based inversions. It also gains a substantial computational advantage over the original model-space ABIC formulation, achieving over a 70-fold speedup in synthetic tests while preserving statistical rigor. This facilitates a smooth transition of ABIC-based MT inversion from 2-D to 3-D and also opens

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the door to its broader application in other large-scale inverse problems. Finally, we briefly consider a field data application of the proposed method to the Yellowstone–Snake River Plain region using EarthScope USArray MT data to showcase its practical applicability.

Keywords: three-dimensional, magnetotelluric, inversion, Akaike's Bayesian Information Criterion, data space variant

INTRODUCTION

Our understanding of the Earth's interior is largely derived from geological and geophysical surveys, including geophysical inversion that aims to determine the plausible spatial distribution of physical properties that match observed data. However, most geophysical inverse problems are inherently ill-posed, making the solutions non-unique and unstable (Zhdanov, 2002; 2015). Regularization techniques, particularly Tikhonov regularization (Tikhonov and Arsenin, 1977), are widely applied to address this challenge in our field.

In Tikhonov-regularized inversion, the objective function to be minimized typically consists of a weighted sum of the data misfit term and the model regularization term. In practice, particularly when reliable prior subsurface information is lacking, a smoothness regularization is commonly adopted (e.g., Constable et al., 1987). The smoothness of the resulting model is controlled by the regularization parameter, also known as the trade-off parameter. This parameter plays a critical role in determining the final solution (e.g., Li and Oldenburg, 1999; Zhdanov, 2002). When it is too small, the ill-posed nature of the problem persists, leading to an unstable inversion process and

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an overly rough solution. On the other hand, if the regularization parameter is too large, the resulting model may be too smooth and fail to sufficiently fit the observed data (Zhdanov, 2002, 2015).

Determining the appropriate regularization parameter that controls the overall smoothness of the model is a challenging task due to the ill-posed nature of geophysical inversions. In two-dimensional (2-D) and three-dimensional (3-D) magnetotelluric (MT) inversion studies, smoothness is often determined through trial-and-error methods (e.g., Rodi and Mackie, 2001) or cooling-type strategies (Zhang et al., 2012; Egbert and Kelbert, 2012; Kelbert et al., 2014). These methods are simple to implement and somewhat reasonable, but lack objectivity. Taking the cooling approach as an example, its underlying philosophy aligns with geological reasoning, where an inversion procedure progressively refines broader, regional structures into more detailed ones. This mirrors the natural process of uncovering intricate geological features (e.g., Miensopust, 2017; Ren and Kalscheuer, 2020). However, the cooling approach can sometimes introduce bias, as all of its settings—such as the initial value of the regularization parameter, the decreasing rate, and the minimum threshold—are based on subjective judgments. These choices often lack a solid quantitative explanation, which can lead to solutions influenced more by the assumptions built into the process. Consequently, while geologically intuitive, the cooling approach can be sensitive to user-defined parameters, potentially leading to less objective outcomes.

The L-curve criterion is another widely used method for determining regularization parameters in our field, involving a plot that captures the trade-off between data misfit and model regularization terms (e.g., Li and Oldenburg, 1999; Farquharson and Oldenburg, 2004; Usui et al.,

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2024; Diba et al., 2025; Song et al., 2025). Typically L-shaped, its "knee point"—where curvature peaks—is deemed a relatively optimal choice. Though rooted in solid mathematical theory (e.g., Hansen, 1992), its geophysical inversion applications suffer notable subjectivity and limitations (e.g., Constable et al., 2015): manual definition of the parameter range for curve generation introduces bias, while the knee point selection is confined to discrete values within this range—often limited in number due to the high computational cost of 3-D MT inversion. Moreover, practical 3-D MT studies (e.g., Usui et al., 2024) rarely exhibit the sharp corner predicted by idealized formulations (e.g., Hansen and O'leary, 1993); instead, a broad curvature with vague maximum curvature makes the result better interpreted as a range of acceptable values rather than a single optimum. And when selecting what one deems the optimal point from this range, personal subjective judgment comes into play (Song et al., 2025).

Constable et al. (1987) incorporated Occam's razor into MT inversion, developing a method for selecting the regularization parameter that yields a model with a data misfit matching the predefined target while minimizing structural complexity. With its clear philosophy and convenient implementation, the Occam inversion and its variants quickly gained widespread attention in the electromagnetic (EM) research community. Over time, it has been extensively adopted and refined for 2-D (e.g., deGroot-Hedlin and Constable, 1990; Siripunvaraporn and Egbert, 2000, 2007; Egbert, 2012) and also 3-D (e.g., Siripunvaraporn and Sarakorn, 2011) MT inversion. However, as many studies have highlighted (e.g., Li and Oldenburg, 1999; Farquharson and Oldenburg, 2004; Constable et al., 2015), the Occam's inversion method and its variants are highly dependent on accurately estimating the uncertainty in the observed data, specifically the

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standard deviation of the MT response function, and the accuracy of the forward calculation. When the noise level is unknown or poorly estimated, or when forward modeling inaccuracies are not properly accounted for (e.g., Baba, 2023), these methods become prone to underfitting or overfitting.

Moreover, as semi-continental and continental MT surveys have gradually matured in recent years (e.g., Dong et al., 2013; Thiel et al., 2020; Schultz et al., 2025), and as the demand for improved understanding of various geological processes has grown (e.g., Usui et al., 2021; Finn et al., 2022; Bennington et al., 2025), the community has placed increasing demands on the resolution and scalability of MT imaging. This shifting landscape has driven advancements across multiple facets of 3-D MT inversion. Key developments include flexible meshing strategies—such as unstructured tetrahedral elements (e.g., Usui, 2015; Usui et al., 2018; Ansari and Craven, 2023; Chen et al., 2025) and deformed hexahedral elements (e.g., Kordy et al., 2016; Munch and Grayver, 2023; Usui et al., 2024)—which allow accurate representation of rugged topography and coastlines, thereby mitigating distortions from surface topographical relief and coastal effects; the development of efficient data-space inversion algorithms to manage the computational demands of large-scale surveys (e.g., Siripunvaraporn and Sarakorn, 2011; Egbert, 2012; Kordy et al., 2016; Usui et al., 2017); and the integration of high-performance computing into 3-D forward and inversion modelling (e.g., Dong et al., 2024; Varılsüha, 2024).

Despite these advances, both emerging and well-established 3-D inversion schemes continue to rely heavily on subjective decisions when selecting the regularization parameter that ultimately controls model smoothness. At present, this setting is typically determined through extensive trial-

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and-error procedures across multiple inversion runs, in the absence of a consistent and objective framework. This subjectivity not only increases solution non-uniqueness, hampers reproducibility, and introduces interpretational bias, but also substantially elevates computational costs—issues that are particularly pronounced in large-scale MT studies.

As such, a theoretically grounded, relatively objective, and computationally tractable inversion framework—one that minimizes reliance on manual regularization parameter tuning and conserves computational resources—is essential for enhancing the repeatability, objectivity, and overall robustness of 3-D MT inversion outcomes, benefiting both the research community and industry. Addressing this need constitutes the central focus of the present study.

In this work, we provide a feasible solution from a statistical perspective to address the challenge of determining the regularization parameter, and thereby controlling the smoothness of the final model, in 3-D MT inversion. Based on Bayes' theorem, we treat the model smoothness term as a prior distribution of the model parameters (Jackson and Matsu'ura, 1985). The Bayesian marginal likelihood function, which quantifies how well the model under the given hyperparameter explains the observed data, is then constructed by combining two probability density functions associated with data misfit minimization and model roughness minimization. We derive the model that maximizes the Bayesian marginal likelihood, with the optimal regularization parameter and model smoothness level determined in this process.

To evaluate marginal likelihood in a numerically stable manner, we adopt Akaike's Bayesian Information Criterion (ABIC; Akaike, 1980), where a smaller ABIC value indicates a higher marginal likelihood, statistically favoring a better hyperparameter, i.e., the regularization

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parameter in this study. Benefiting from its solid theoretical foundation and near-complete objectivity in practical applications, ABIC has gained traction in 2-D and 3-D EM inversions (e.g., Uchida, 1993; Ogawa and Uchida, 1996; Mitsuhata et al., 2002; Mitsuhata and Uchida, 2002; Uchida and Sasaki, 2006) and other geophysical inversions (e.g., Yabuki and Matsu'ura, 1992; Murata, 1993; Fukahata et al., 2003, 2004; Honsho et al., 2012; Tsuyuki et al., 2021).

However, despite its advantages and application potential, this statistical approach has not yet seen widespread application in 3-D MT or other large-scale inverse problems, primarily due to the associated computational burdens (e.g., Honsho et al., 2012). To further overcome this limitation and enhance its practical usability, we propose a data-space variant of ABIC (DABIC), incorporating a low-rank update approach akin to model-to-data space inversion (e.g., Siripunvaraporn and Egbert, 2000; Siripunvaraporn et al., 2005; Usui et al., 2017). This data-space formulation of ABIC significantly improves computational efficiency while preserving statistical rigor, thereby enabling the smooth extension of ABIC-based inversion from 2-D to 3-D and supporting its broader application to large-scale MT and other inverse problems beyond MT. Proposing and verifying this new tractable formulation of ABIC in this work represents a key step toward bridging the theoretical strengths of this statistical approach and the practical demands of 3-D MT inversion and other large-scale inverse studies.

We begin our detailed discussion with a theoretical analysis of the proposed inversion method, including the assumptions and mathematical background of this statistical framework, the detailed derivation of ABIC and DABIC, and the procedure for obtaining maximum marginal likelihood solutions in 3-D MT inversion. We then evaluate the performance of the proposed method through

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synthetic inversions, assessing stability under different inversion configurations. Additionally, by benchmarking against other inversions—including inversions based on the L-curve criterion, the cooling approach, and the original ABIC formulation in model space—we demonstrate the objectivity and computational efficiency of the proposed method. Finally, we briefly apply this method to the Yellowstone–Snake River Plain region using EarthScope USArray MT data (Schultz et al., 2025), comparing the result with previous studies (Kelbert et al., 2012; Meqbel et al., 2014) to assess its practical applicability.

THEORY

Objective function and non-linear regression

Geophysical inversion aims to determine the plausible spatial distribution of physical properties that fit observed data. In 3-D MT inversion, the unknown model parameters typically represent the Earth's electrical resistivity or some functions thereof; these are modeled as a collection of polyhedra (Zhdanov, 2002) and defined by an M-dimensional model vector $\mathbf{m}$. When the observed data are represented by an N-dimensional vector $\mathbf{d}$, the goodness-of-fit between the observed and calculated data can be expressed using the following L_2-norm measure:

$$S = \left\| \mathbf{Wd} - \mathbf{WF}(\mathbf{m}) \right\|^2 \tag{1}$$

where $\|\cdot\|$ denotes the L_2-norm; $\mathbf{F}(\mathbf{m}) \in ¡^{N}$ denotes the forward calculated data vector, and the physical system governing the relationship between $\mathbf{m}$ and $\mathbf{F}(\mathbf{m})$ is governed by Maxwell's equations (e.g., Chave and Jones, 2012). $\mathbf{W} \in ¡^{N \times N}$ is the data weighting matrix, when the data

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noise is uncorrelated, **W** is a diagonal matrix, whose diagonal elements are the reciprocals of the standard deviations ($v_{r,i}$) of the MT response function：

$$w_{ii} = \frac{1}{v_{r,i}} \quad (i = 1, 2, \ldots, N) \tag{2}$$

Since the absolute magnitude of data errors is often difficult to estimate in practice, the data weights used here should be interpreted as relative weights (e.g., Mitsuhata and Uchida, 2002; Farquharson and Oldenburg, 2004).

Due to severe ill-posedness, minimizing Equation 1 using iterative methods like Gauss-Newton optimization often fails to produce a realistic subsurface model. A regularization functional, often a smoothness stabilizer (e.g., Siripunvaraporn et al., 2005; Kelbert et al., 2014; Usui, 2015), is thus crucial in mitigating the ill-posedness during inversion. The objective function corresponding to the regularized inversion can be expressed as:

$$U(\mathbf{m}) = \left\|\mathbf{Wd} - \mathbf{WF(m)}\right\|^2 + \alpha^2 \left\|\mathbf{Cm}\right\|^2 \tag{3}$$

Here, α is the regularization parameter, and **C** is the roughness operator, which calculates the gradient of the model parameters between laterally and vertically adjacent cells. We can employ different orders of difference filters as roughness operators (e.g., deGroot-Hedlin and Constable, 1990; Mitsuhata, 2004). In this study, the first-order operator is adopted to explore the feasibility of the proposed method.

Consider the linear Taylor approximation of the data for the model to be found at the (k+1)-th iteration:

$$\mathbf{F}(\mathbf{m}_{k+1}) \approx \mathbf{F}(\mathbf{m}_k) + \mathbf{J}_k(\mathbf{m}_{k+1} - \mathbf{m}_k) \tag{4}$$

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where $\mathbf{m}_k$ and $\mathbf{m}_{k+1}$ represent the k-th and (k+1)-th model vectors, respectively, with the default initial value of k being 0; $\mathbf{J}_k \in \mathbb{R}^{N\times M}$ is the sensitivity matrix corresponding to the k-th model $\mathbf{m}_k$.

The Gauss-Newton optimization method is adopted to obtain the resistivity structure that minimizes the objective function iteratively (e.g., Usui, 2015):

$$\left[\left(\mathbf{W}\mathbf{J}_k\right)^T\left(\mathbf{W}\mathbf{J}_k\right)+\alpha^2\mathbf{C}^T\mathbf{C}\right]\delta\mathbf{m}_{k+1}=\hat{\mathbf{d}}_k \tag{5}$$

$$\hat{\mathbf{d}}_k=-\mathbf{C}^T\mathbf{C}\mathbf{m}_k+\left(\mathbf{W}\mathbf{J}_k\right)^T\mathbf{W}\left(\mathbf{d}-\mathbf{F}(\mathbf{m}_k)\right) \tag{6}$$

$\delta\mathbf{m}_{k+1}$ denotes the model increment based on the k-th model, which is obtained in the k-th iteration by solving Equation 5. Furthermore, in 3-D MT inversion, the number of data points is generally smaller than that of model parameters. As such, shifting inversion from the model space to the data space has become a well-recognized strategy for enhancing computational efficiency (e.g., Siripunvaraporn et al., 2005; Kordy et al., 2016; Usui et al., 2017). A feasible method to realize this transformation is to update the coefficient matrix in Equation 5 via the Sherman–Morrison–Woodbury formula (Golub and Van Loan, 2013):

$$\left[\left(\mathbf{W}\mathbf{J}_k\right)^T\left(\mathbf{W}\mathbf{J}_k\right)+\alpha^2\mathbf{C}^T\mathbf{C}\right]^{-1}=\left(\alpha^2\mathbf{C}^T\mathbf{C}\right)^{-1}-\left(\alpha^2\mathbf{C}^T\mathbf{C}\right)^{-1}\left(\mathbf{W}\mathbf{J}_k\right)^T\boldsymbol{\Gamma}_k^{-1}\left(\mathbf{W}\mathbf{J}_k\right)\left(\alpha^2\mathbf{C}^T\mathbf{C}\right)^{-1} \tag{7}$$

$$\boldsymbol{\Gamma}_k=\mathbf{I}+\left(\mathbf{W}\mathbf{J}_k\right)^T\left(\alpha^2\mathbf{C}^T\mathbf{C}\right)^{-1}\left(\mathbf{W}\mathbf{J}_k\right) \tag{8}$$

After this transformation, the dimension of the dense target matrix to be inverted shifts from the number of model parameters shown in Equation 5 to the number of data points expressed in Equation 8. This shifting in the dimension of the inverse matrix has led to such methods being termed data space inversions (e.g., Usui et al., 2017).

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However, regardless of the inversion method employed, the regularization parameter serves as a trade-off between data misfit and model roughness. Beyond its role as a formal balance between these two terms, it also exerts crucial control over inversion stability and the shaping of the solution space. Given that inversion results are highly sensitive to the value of this parameter, careful selection of its magnitude is imperative—and this is precisely the core focus of the present study.

Bayesian statistics

Applying Bayesian statistics to the inversion objective function expressed in Equation 3, we treat the data-fitting term as the data distribution and the model constraint term as a prior distribution. Suppose the difference between each set of the weighted observed and modeled responses follows a Gaussian distribution with zero mean and an unknown constant variance v_0^2. The probability density function (PDF) of the weighted data **Wd**, given model **m**, can be expressed as follows:

$$f(\mathbf{Wd} \mid \mathbf{m}, v_0^2) = \left(2\pi v_0^2\right)^{-N/2} \exp\left(-\frac{\|\mathbf{Wd - WF(m)}\|^2}{2v_0^2}\right) \tag{9}$$

We assume that the prior model distribution also follows a Gaussian distribution. Since the variance of this Gaussian distribution is unknown, we adopt the assumption from Ogawa and Uchida (1996) that it takes the form of v_0^2/α^2. The PDF for the prior model can then be expressed as:

$$p(\mathbf{m} \mid v_0^2, \alpha^2) = \left(2\pi v_0^2\right)^{-M/2} \left|\alpha^2 \mathbf{C}^T \mathbf{C}\right|^{\frac{1}{2}} \exp\left(-\frac{\alpha^2 \|\mathbf{Cm}\|^2}{2v_0^2}\right) \tag{10}$$

where $|\cdot|$ denotes the determinant of a matrix. In Uchida (1993), the rank of the matrix in this

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determinant calculation is $M-1$, whereas it is M in this work after regularizing the matrix, which we will discuss later.

According to Bayes' theorem (e.g., Aster et al., 2019), the posterior probability density function of the model parameters **m** given the observed data weighted by data weighting matrix **Wd** can be expressed as:

$$q(\mathbf{m}\,|\,\mathbf{Wd},v_0^2,\alpha^2)=\frac{f(\mathbf{Wd}\,|\,\mathbf{m},v_0^2)\,p(\mathbf{m}\,|\,v_0^2,\alpha^2)}{L(\mathbf{Wd}\,|\,v_0^2,\alpha^2)} \tag{11}$$

Here, L is the marginal likelihood (also known as the model evidence), and it is no longer a function of the model **m**. By integrating Equation 11 over the entire model space and noting that the PDF q integrates to one over this space, we obtain the expression of L:

$$\begin{aligned} L(\mathbf{Wd}\,|\,v_0^2,\alpha^2)&=\int f(\mathbf{Wd}\,|\,\mathbf{m},v_0^2)\,p(\mathbf{m}\,|\,v_0^2,\alpha^2)d\mathbf{m}\\ &=\int\left(2\pi v_0^2\right)^{-(N+M)/2}\left|\alpha^2\mathbf{C}^T\mathbf{C}\right|^{\frac{1}{2}}\times\exp\left(-\frac{1}{2v_0^2}\left(\left\|\mathbf{Wd}-\mathbf{WF(m)}\right\|^2+\alpha^2\left\|\mathbf{Cm}\right\|^2\right)\right)d\mathbf{m} \end{aligned} \tag{12}$$

L quantifies how well the model, under the given set of hyperparameters v_0^2, α^2, explains the weighted observed data, averaging over all plausible model vectors **m** weighted by their prior probabilities, and therefore its maximization enables the statistically optimal set of hyperparameters selection.

However, since the integral in Equation 12 is analytically intractable for nonlinear inversion, we first reformulate it through linearized approximation, thereby facilitating numerical maximum likelihood estimation. The term within the parentheses of the exponential is associated with the objective function U. To perform the integration on the right-hand side, we approximate it by

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substituting Equation 4 into Equation 12, and then employ the Gaussian integral (e.g., Equation B-3 in Mitsuhata et al., 2002), yielding the following expression:

$$L_{k+1}^{l}(\mathbf{Wd}\,|\,v_0^2,\alpha^2)=\left(2\pi v_0^2\right)^{-N/2}\left|\alpha^2\mathbf{C}^T\mathbf{C}\right|^{\frac{1}{2}}\left|\left(\mathbf{WJ}_k\right)^T\mathbf{WJ}_k+\alpha^2\mathbf{C}^T\mathbf{C}\right|^{-\frac{1}{2}}\exp\left(-\frac{U_l(\mathbf{m}_{k+1})}{2v_0^2}\right) \quad (13)$$

where

$$U_l(\mathbf{m}_{k+1})=\left\|\mathbf{Wd}-\mathbf{W}\left(\mathbf{F}(\mathbf{m}_k)+\mathbf{J}_k(\mathbf{m}_{k+1}-\mathbf{m}_k)\right)\right\|^2+\alpha^2\left\|\mathbf{Cm}_{k+1}\right\|^2 \quad (14)$$

In Equation 13, among the two hyperparameters, v_0^2 that maximizes the linearized marginal likelihood can be determined analytically via its derivative, whereas α^2 does not admit an analytical derivative. Therefore, numerical optimization methods (e.g., line search) are required to determine the value of α^2 that maximizes the marginal likelihood.

We first choose hyperparameter v_0^2 so that L is maximized by differentiating Equation 13 with respect to v_0^2 and then setting the derivative to zero.

$$\frac{\partial L_{k+1}^{l}(\mathbf{Wd}\,|\,v_0^2,\alpha^2)}{\partial v_0^2}=0 \quad (15)$$

The linearized marginal likelihood in Equation 13 reaches its maximum when the following condition is satisfied:

$$v_0^2=U_l(\mathbf{m}_{k+1})/N \quad (16)$$

We substitute Equation 16 into Equation 13:

$$L_{k+1}^{l}(\mathbf{Wd}\,|\,\alpha^2)=\left(2\pi U_l(\mathbf{m}_{k+1})/N\right)^{-N/2}\left|\alpha^2\mathbf{C}^T\mathbf{C}\right|^{\frac{1}{2}}\left|\left(\mathbf{WJ}_k\right)^T\mathbf{WJ}_k+\alpha^2\mathbf{C}^T\mathbf{C}\right|^{-\frac{1}{2}}\exp\left(-\frac{N}{2}\right) \quad (17)$$

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To this point, one can observe that the linearized marginal likelihood depends only on α^2. Maximizing it through a numerical approach can directly provide a statistically optimal solution for regularization parameter selection, as Ogawa and Uchida (1996) did. For numerical stability and convenience, however, it is customary to work with its logarithmic form ABIC (e.g., Akaike, 1980; Uchida, 1993; Mitsuhata and Uchida, 2002).

ABIC and its data-space variant

Since the ABIC indicator is the negative logarithmic form of the marginal likelihood function, the maximum of the marginal likelihood function shown in Equation 17 corresponds to the minimum of ABIC (Akaike, 1980):

$$\mathrm{ABIC} = (-2)\log(L(\mathbf{Wd}\,|\,v_0^2,\alpha^2)) \;+\; 2\dim(\text{hyperparameters}) \tag{18}$$

Since only the regularization parameter is the unknown hyperparameter at this stage, the dimensionality of the hyperparameter space is 1.

Here, we adopt the linearized form of the marginal likelihood function shown in Equation 17, and obtain the expression for linearized approximated ABIC as a function of the regularization parameter:

$$\mathrm{ABIC}_{k+1}^{l}(\alpha) = N\log\left(2\pi\frac{U_l(\mathbf{m}_{k+1})}{N}\right) - \log\left|\alpha^2\mathbf{C}^T\mathbf{C}\right| + \log\left|\left(\mathbf{WJ}_k\right)^T\mathbf{WJ}_k + \alpha^2\mathbf{C}^T\mathbf{C}\right| + N + 2 \tag{19}$$

With the same statistical meaning as the marginal likelihood discussed previously, the ABIC quantifies how well the model, under a given regularization parameter, explains the weighted observed data by averaging over all plausible models weighted by their prior probabilities. A smaller ABIC value, corresponding to a larger marginal likelihood function in Equation 17,

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indicates that the model under the specified regularization parameter better explains the weighted observed data. Therefore, minimizing the ABIC enables the selection of the statistically optimal regularization parameter.

While the linearized expression of ABIC given in Equation 19 provides a practical approach for nonlinear MT inversion, it often fails to achieve an ideal minimum in practice due to linearization errors, as noted in Mitsuhata and Uchida (2002). Therefore, following Uchida (1993) and Mitsuhata et al. (2002), we compute the exact objective value $U(\mathbf{m}_{k+1})$ through Equation 3 and substitute it for $U_l(\mathbf{m}_{k+1})$ in the ABIC evaluation:

$$\mathrm{ABIC}_{k+1}(\alpha) = N \log\left(2\pi \frac{U(\mathbf{m}_{k+1})}{N}\right) - \log\left|\alpha^2 \mathbf{C}^T\mathbf{C}\right| + \log\left|\left(\mathbf{W}\mathbf{J}_k\right)^T \mathbf{W}\mathbf{J}_k + \alpha^2 \mathbf{C}^T\mathbf{C}\right| + N + 2 \quad (20)$$

The ABIC formulation in Equation 20 has been successfully applied to nonlinear EM inverse problems for 2-D (e.g., Uchida, 1993), 2.5-D (e.g., Mitsuhata et al., 2002), and 3-D settings (Uchida and Sasaki, 2006). We extended this approach directly to 3-D MT inversion as well. However, the dramatic increase in the number of model parameters in 3-D problems leads to a prohibitive computational cost, posing a major challenge for practical application and compelling us to seek a more flexible and computationally efficient way for evaluating ABIC to maximize the marginal likelihood.

As shown in Equation 20, the calculation of ABIC involves the determinant of two matrices (the second and third terms on the right-hand side). For the second term, due to the sparsity of the roughness matrix, high-performance computing platforms such as PARDISO (with contributions from Pasadakis et al., 2023; Eftekhari et al., 2021; Gaedke-Merzhäuser et al., 2023) can be

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employed for efficient computation (e.g., Kordy et al., 2016; Usui et al., 2017). However, for the third term on the right-hand side, the target matrix is dense, and calculating its determinant, like solving the inverse of this matrix, is computationally expensive. The computational complexity scales cubically with the matrix dimension M (Golub and Van Loan, 2013), the number of model parameters. In 3-D MT inversion, M can easily exceed several tens of thousands, and this immense computational cost poses a significant challenge to the practical application of ABIC in 3-D MT and other large-scale surveys.

To address this issue, we leverage the concept of low-rank updates in matrix determinant calculations (Equation A10 in Rasmussen and Williams, 2006) and transform the determinant computation from the model space to the data space:

$$\left|\left(\mathbf{W}\mathbf{J}_k\right)^T \mathbf{W}\mathbf{J}_k + \alpha^2\mathbf{C}^T\mathbf{C}\right| = \left|\alpha^2\mathbf{C}^T\mathbf{C}\right| * \left|\mathbf{I} + \mathbf{W}\mathbf{J}_k\left(\alpha^2\mathbf{C}^T\mathbf{C}\right)^{-1}\left(\mathbf{W}\mathbf{J}_k\right)^T\right| \tag{21}$$

This approach is analogous to the low-rank update idea used during matrix inversion in the transition from model space inversion to data space inversion. A prerequisite is that the roughness matrix $\mathbf{C}$ must be nonsingular. While in practice the roughness matrix is not of full rank (Uchida, 1993), $\mathbf{C}$ can be regularized by adding a small positive value to its diagonals, as proposed by Kordy et al. (2016). Our tests confirm that this regularization renders the roughness matrix nonsingular, and when the positive value is sufficiently small (e.g., 1.0×10^{-4}), its influence on the results is negligibly small. After this transformation, we arrive at the expression for DABIC—a data-space variant of ABIC indicator:

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$$\mathrm{DABIC}_{k+1}(\alpha) = N \log\left(2\pi \frac{U(\mathbf{m}_{k+1})}{N}\right) + \log\left|\mathbf{I} + \mathbf{W}\mathbf{J}_k \left(\alpha^2 \mathbf{C}^T \mathbf{C}\right)^{-1} \left(\mathbf{W}\mathbf{J}_k\right)^T\right| + N + 2 \tag{22}$$

We perform matrix decomposition using LAPACK (Anderson et al., 2000) and compute the matrix's determinant accordingly. It can be seen that after the transformation from Equation 20 to Equation 22, the dimensionality of the dense matrix determinant calculation shifts from the number of model parameters to the number of data points. In most 3-D MT inversions, this can significantly save computational costs.

Furthermore, the dense target matrix used for determinant calculation in Equation 22 is identical to the matrix obtained in the data-space inversion presented in Equation 8. This implies that the matrix decomposition used for matrix inversion can be directly utilized in the determinant calculation. As a result, applying the DABIC proposed in this work introduces almost no additional computational burden to data-space inversion; we refer to the data-space inversion based on the DABIC indicator as the D-DABIC method hereinafter.

Minimization of DABIC (maximization of marginal likelihood)

In each linearized iteration, we first narrow down the approximate interval where the minimum of DABIC lies, and then use Brent's line search method (Brent, 1973), which combines the stability of the golden-section search used in Occam's inversion (Constable et al., 1987) with the efficiency of quadratic line search, to find the regularization parameter that minimizes the DABIC, along with the corresponding updated model vector.

Inversion termination criteria

We have designed two types of iteration termination criteria for the inversion: one associated

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with step size, and the other with data fitting. Both are elaborated in detail below.

For the step size-related termination criteria, the mechanism works as follows: At the start of each line search, an initial step size (e.g., 1.0) is assigned. If Brent's line search gives a model where both the data misfit and DABIC are reduced compared to their values in the previous iteration, the search for the current step size is considered successful, and the algorithm advances to the next iteration. If the minimum DABIC identified by Brent's line search remains higher than that of the previous iteration, or if there is no improvement in data fitting, the step size will be reduced (e.g., halved) and Brent's line search will be restarted. This process is repeated iteratively. If the step size drops below a threshold (e.g., 0.1) or the number of step size reductions within a single iteration reaches an upper limit (e.g., 3 times), and a suitable new model is still not found, the inversion result from the previous iteration is considered to have sufficiently approached the desired solution, and the inversion is terminated. This strategy is similar to that employed in 2-D Occam's inversion by deGroot-Hedlin and Constable (1990). However, given the significantly higher computational cost of 3-D forward and inverse calculation, we use fewer retrial attempts (3 versus 7).

Turning to the data fitting-related termination criteria: When data uncertainty is accurately known, the inversion is considered to have adequately fit the data once the Root Mean Square (RMS) error, as defined by Equation 23, converges to 1.0. In cases where data uncertainty is not accurately known, if the decreasing rates of the data misfit terms (defined by Equation 24) are all smaller than a threshold (e.g., 0.01) for multiple consecutive iterations (e.g., 3 iterations), it is reasonable to conclude that the inversion model has approached the expected solution, and the

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inversion is terminated.

$$\mathrm{RMS} = \sqrt{\frac{S}{N}} = \sqrt{\frac{1}{N}\left\|\mathbf{Wd - WF(m)}\right\|^2} \tag{23}$$

$$\varepsilon_{k+1} = \frac{\left(S_{k+1} - S_k\right)}{S_k} = \frac{\left\|\mathbf{Wd - WF(m}_{k+1}\mathbf{)}\right\|^2 - \left\|\mathbf{Wd - WF(m}_k\mathbf{)}\right\|^2}{\left\|\mathbf{Wd - WF(m}_k\mathbf{)}\right\|^2} \tag{24}$$

The overall workflow of the D-DABIC inversion method is summarized in the form of a flowchart as shown in Figure 1.

NUMERICAL EXPERIMENTS

Synthetic model configuration

Presented here are inversion verifications using synthetic MT data. We simulate a 3-D MT inversion scenario where the MT stations in the study area include both densely spaced stations along a key profile and sparsely distributed stations for regional study. The underground electrical resistivity structure is designed mainly based on the structure around the Atotsugawa Fault system in central Japan imaged by Usui et al. (2021). This simplified model captures the main structures identified in the previous research, which are characterized by moderate to high resistivity in the upper to middle crust (from the surface to approximately 10 km depth) and two sporadically distributed localized high-conductivity anomalies. Additionally, it includes three prominent high-conductivity anomalies in the mid to lower crust, which are attributed to fluid-rich zones.

The vertical cross-section (V-CS, 1st row) and horizontal cross-sections (H-CSs, 2nd row) of this model are presented in Figure 2. The V-CS is taken along the dense profile from west to east,

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and the three H-CSs correspond to depths of 1km, 5 km, and 25 km. Thirty-three MT stations are marked with black inverted triangles. Synthetic impedance tensors are calculated for 10 periods, ranging from 0.1 to 10000 seconds, by solving Maxwell's equations on a 34 × 34 × 44 grid. Under this configuration, the total number of data points N = 33(station) × 10(frequency) × 8(impedance) = 2640, and the total number of model parameters M = 50864.

Performance under varying settings

We first evaluated the performance of the proposed D-DABIC inversion method under different noise levels. Three tests were conducted, where Gaussian random noise with standard deviations of 2%, 3%, and 5% of the maximum amplitude of the off-diagonal impedance tensor components (i.e., $max(|Zxy|, |Zyx|)$) was added to the data. Convergence was achieved after 7, 6, and 6 iterations, respectively, and the final iteration of each case was used to verify convergence. The resulting models are shown in Figure 3.

To evaluate the accuracy of the inverted models in reproducing the true synthetic model, we quantified the model misfit using the mean absolute difference of model parameters ($\log_{10}$ resistivity), as defined in Equation 25, where m_i^{true} and $m_i^{inverted}$ denote the i-th true synthetic model and inverted model parameter, respectively. The model misfits are 0.117, 0.120, and 0.149 for inversions using data with 2%, 3%, and 5% noise, respectively, indicating that the structures are less well-resolved as higher levels of noise are added. This is expected, as the increased noise obscures the realistic signals to a greater extent. Therefore, although the data of all tests converged, the fitted data is influenced by varying degrees of noise, causing the solution to deviate from the true solution to different extents.

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$$model\ misfit = \frac{1}{M}\sum_{i=1}^{M}\sqrt{(m_i^{true} - m_i^{inverted})^2} \tag{25}$$

$$model\ modfication = \frac{1}{M}\sum_{i=1}^{M}\sqrt{(m_i^{(k+1)} - m_i^{(k)})^2} \tag{26}$$

Key inversion parameters, such as the selected regularization parameter that minimizes DABIC, the absolute mean model modification, DABIC value, model roughness term, and RMS misfit as functions of iteration, are shown in Figure 4. The absolute mean model modification is expressed in Equation 26, where $m_i^{(k+1)}$ and $m_i^{(k)}$ denote the i-th model parameter in the $(k+1)$-th and k-th iterations, respectively. Since the initial and final values of the regularization parameter and RMS misfit differ significantly, we use a logarithmic scale to illustrate their variations.

During the initial stage of inversion, the initial model (i.e., a homogeneous half-space) leads to significant discrepancies between the calculated responses and observed data (Figure 4e). This results in a large space for model updates and severe non-uniqueness. To prevent overly large model updates that may produce unrealistic structures, the D-DABIC inversion selects a larger regularization parameter automatically to enhance the stability of the inverse problem and promote a smoother solution. As the inversion proceeds, the data residuals gradually decrease (Figure 4e), reducing the potential for large model updates (Figure 4d). Consequently, the regularization parameter that minimizes DABIC shows an overall decreasing trend (Figure 4a), allowing the model roughness to increase progressively (Figure 4c) to ensure adequate data fitting.

The regularization parameters that determine the smoothness of the solution derived from minimizing DABIC, as developed in this study, and those obtained by performing a line search to

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find one that minimizes data RMS misfit in each iteration, as described in Occam's inversion method (Figure 3 in Constable et al., 1987), both exhibit an overall decreasing trend with iteration. Figure 5 shows DABIC and data RMS as functions of the regularization parameter for the inversion case containing 3% Gaussian random noise. In the initial iterations, the minimum points of DABIC and data RMS misfit are distinct and nearly coincident. However, in the later stages of the iteration process, the regularization parameter that minimizes data misfit is generally smaller than the one that minimizes DABIC. For instance, during the 4th and 5th iterations, the regularization parameters tested in the line search that yielded the lowest RMS were 0.75 and 0.59, respectively, while those that minimized DABIC were 0.8 and 0.7. This discrepancy arises because DABIC not only accounts for data fitting but also model roughness and the properties of the matrix in Equation 22. A small regularization parameter increases the determinant of the target matrix in Equation 22, while simultaneously, a rough model contributes to an increase in the model roughness term in the objective function U.

We further analyze the data fitting results, as shown in Figures 6a, 6b, and 6c, which present the final RMS misfit across all frequencies for each station under three noise levels. Overall, the inversion fittings under all three noise levels are relatively homogeneous, with RMS values mostly ranging between 0.75 and 1.5. No regions of obvious overfitting or underfitting are observed. We also examine the systematic bias in fit as a function of frequency, as discussed by Smith and Booker (1988) and deGroot-Hedlin and Constable (1990). They describe cases with nearly uniform fitting across the frequency band as "white" fits, while uneven fitting—such as overfitting in the low-frequency range—is referred to as "red" fits, and overfitting in the high-frequency range is

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called “blue” fits.

In cases of uneven fitting (e.g., Smith and Booker, 1988), even if the overall RMS converges to an ideal range, overfitting in certain frequency bands and underfitting in others can lead to the loss of significant structure. The plots in Figures 6e and 6f show that the weighted residuals of two randomly selected stations are essentially uncorrelated with frequency for all three noise levels. Thus, it appears that the inversion method and the corresponding inversion configurations specified here—such as frequency range, mesh discretization, and station distribution—do not introduce obvious systematic bias issues.

The objective function U may have multiple local minima, which could cause the inversion model to depend on the choice of the initial model. While trial and error with different initial models can provide a general sense of this dependency, it is impossible to completely rule out the possibility that an untested initial model might lead to a different solution. However, if the inversion consistently converges to the same or a very similar model, regardless of whether it starts from a rough model or a homogeneous half-space, one can be more confident that the globally optimal model has been achieved. To verify this, we performed D-DABIC using three different initial models: two uniform half-spaces with resistivities of 10 Ω·m and 1000 Ω·m, and a rough model. All inversions used data with 3% noise. As shown in Figure 7, the resulting models are highly consistent with each other and with the one obtained using a 100 Ω·m half-space as the initial model discussed earlier. This suggests that the influence of the initial model on this D-DABIC method is minimal.

Comparison with several other inversions

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Here, we compare D-DABIC with several other inversion schemes: data-space inversion with model-space ABIC (D-MABIC), data-space inversion utilizing an empirical L-curve method with fixed regularization parameters for separate inversions (e.g., Usui et al., 2024; D-Lcurve), and data-space inversion employing a cooling scheme (D-Cooling) to determine the regularization parameter in each iteration and final model. We have not included the Occam inversion method or its variants in this comparison, as their advantages and drawbacks have been thoroughly analyzed and studied in previous works (e.g., Constable et al., 2015). All inversion runs used the same starting model (a 100 Ω·m half-space) and the identical mesh discretization discussed in the previous section and were performed on the same high-performance computing platform: an AMD EPYC 9654 96-Core Processor, with 10 MPI processes and 4 OpenMP threads utilized for each task. The inversion is performed using data generated in the previous section, which contains Gaussian Random noise with a standard deviation of 3% of the maximum amplitude of the off-diagonal impedance tensor components.

Overall, all the investigated inversion methods reconstruct the main features of the model and produce responses that fit the data adequately. Figure 8 shows the final models produced by D-DABIC, D-MABIC, D-Lcurve, and D-Cooling. Although the synthetic model and the solution of D-DABIC have been presented earlier, we include them here again for a more intuitive comparison.

Figure 9 presents several inversion process metrics, including the RMS misfit as a function of CPU time, the RMS misfit, mean model modification, model roughness, and regularization parameter as functions of iteration; and the L-curve plot used in the D-Lcurve inversion to determine the ‘optimal’ regularization parameter. It is important to note that the CPU times shown

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in Figure 9e for D-Lcurve and D-Cooling correspond to the duration of a single inversion run using the 'optimal' settings. To identify these optimal parameters, 16 and 12 separate trials, partly shown in Figure 10, were conducted for D-Lcurve and D-Cooling, respectively. Meanwhile, a comprehensive comparison of the computation times is provided in Table 1.

The D-MABIC inversion converges after six iterations, yielding a solution shown in Figure 8c that closely approximates the one obtained using D-DABIC shown in Figure 8b. As shown in Figures 8 and 9, both the final inversion result and the inversion process metrics exhibit negligible differences between the two methods. These parameters include the regularization parameters, RMS misfit, and model modifications across iterations, all of which are essentially identical to those observed in the D-DABIC approach.

However, under this synthetic configuration, the computational time for D-MABIC is approximately 70 times longer than that for D-DABIC, with D-MABIC requiring 553.6 CPU hours to converge to the final solution, compared to 7.9 CPU hours for D-DABIC. This discrepancy aligns with the computational disparity reported between model-space Occam and data-space Occam inversions in Siripunvaraporn and Egbert (2000). The significantly longer runtime per iteration in D-MABIC is primarily attributed to the increased computational overhead required for factorizing larger matrices in the model space, with dimension M, as described in Equation 20. This comparison effectively demonstrates that our proposed DABIC retains the original statistical properties of ABIC as described in Akaike (1980) while significantly reducing computational costs. As a result, it enables a more efficient and practical application of this versatile approach to large-scale MT inversions as well as other geophysical inversion investigations, as long as the number

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of model parameters exceeds the number of data points.

For the D-Lcurve and D-Cooling inversions, the solutions exhibit a strong dependence on user-specified parameters, as can be clearly observed in Figure 10. In the case of D-Lcurve, as shown in the top panel of Figure 10, both the range and the discrete values of the regularization parameters used to construct the L-curve critically influence the identification of the 'knee' point and, consequently, the final 'optimal' inversion result. These inherent subjectivity and limitations have been previously noted in studies such as Constable et al. (2015), Song et al. (2024), and Song et al. (2025). In this synthetic experiment, after more than ten trials, we determined that a suitable range for the regularization parameter lies between 0.58 and 5.8, with an empirically selected optimal value of 1.0.

Under this setting, the D-Lcurve inversion converges after 12 iterations, significantly exceeding the number of iterations required by D-DABIC and D-MABIC. Analysis of the intermediate inversion results shown in the mid panel of Figure 10 reveals that, although the final model shown in Figure 8d successfully reproduces the main features of the true model to some extent with a model misfit of 0.166, the inversion process itself exhibits substantial instability. As illustrated in Figure 9c, the model roughness varies in a highly discontinuous and abrupt manner, and large model modifications occur even when improvements in data misfit are negligible. This behavior suggests that the regularization strategy employed in the D-Lcurve inversion fails to effectively stabilize the inversion process or adequately mitigate the ill-posedness of the problem. Consequently, the resulting instability introduces artificial variations in the intermediate models, necessitating additional iterations to gradually rectify these artifacts.

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The D-Cooling inversion also exhibits a significant dependence on user-defined parameters, such as the initial regularization parameter, the decreasing rate of the regularization parameter, and in some cases, the data misfit reduction threshold and the lower bound of the regularization parameter. Variations in these settings can lead to markedly different inversion outcomes, as illustrated in the lower panel of Figure 10. The optimal selection of these parameters is influenced by multiple factors, yet no clear, universally accepted criteria exist for their determination—at least not for key indicators like the initial value and decreasing rate. Furthermore, there is even a lack of qualitative guidance comparable to the "knee" point used in D-L curve inversion.

In this synthetic case, we obtained the final result of D-Cooling shown in Figure 8e after more than ten trials, using an initial regularization parameter of 100 and a decreasing rate of 0.75. Under these settings, the D-Cooling inversion converged after 17 iterations, yielding a model with a model misfit of 0.137. Similar to the D-Lcurve method, D-Cooling successfully recovers the main features of the true model but requires significantly more iterations to achieve convergence compared to D-DABIC. Furthermore, the strong dependence on parameter selection poses a substantial challenge to the practical applicability of methods such as D-Lcurve and D-Cooling. Unlike synthetic tests, real-world problems lack a ‘true’ model as a reference for tuning these user-defined parameters, making their optimal selection even more complex.

Example with field data

We briefly consider an application of D-DABIC inversion to a subset of USArray MT data covering the Yellowstone–Snake River Plain region, as depicted in Figure 11, to demonstrate the practical applicability of this method. This dataset comprises measurements from 98 stations

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shown in Figure 11, indicated by white solid dots, collected between 2006 and 2011, which can be freely accessed via the IRIS Data Management Center (EM Transfer Function Product Query at www.iris.edu/spud/emtf), as documented by Kelbert et al. (2018).

We incorporated the full impedance tensor and vertical transfer functions from all 98 stations, resulting in a total of 15,224 data points used in the inversion. For the impedance tensor, we included 14 frequencies spanning the full period range (about 10–18,000 s). To mitigate the influence of external source contamination, vertical transfer functions were excluded for periods exceeding 10,000 s (e.g., Shimizu et al., 2011). A relative error floor of 5% of $|Z_{xy}Z_{yx}|^{1/2}$ is assigned to impedance tensor components, and a constant error floor of 0.03 is assigned to the vertical transfer functions, consistent with the error floor standards used by Meqbel et al. (2014). The inversion grid comprises 147,456 cells (48 × 64 × 48), with a central horizontal cell size of 20 km. The vertical cell thickness increases with depth, from 100 m at the surface to over 100 km at the model bottom; in the crust, cell thicknesses are typically 1–3 km. We employ unstructured, deformed hexahedral elements (e.g., Usui et al., 2024) to accurately represent the topography of the study area.

Our inversion begins with a uniform half-space model of 100 Ω·m. After 15 iterations (about 1000 CPU hours), the inversion found its solution with an RMS misfit of 1.3, representing a reduction of approximately 96% from the initial value. Statistical analysis of the RMS across all the stations shown in Figure 12a reveals that most high discrepancies are concentrated at a few stations, and Figure 12c indicates that these stations are scattered with limited influence on the overall structure. Additionally, we observe relatively poorer fits for short-period data (Figure 12b),

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likely reflecting the coarse grid's limited power in resolving shallow, fine-scale features, as also mentioned by Meqbel et al. (2014) and Bedrosian and Feucht (2014).

The final result of this work (referred to herein as SRPY-MT-DABIC) is presented in Figure 13, which includes horizontal slices of the recovered resistivity structure at depths of 13 km, 35 km, 55 km, and 120 km, as well as a vertical cross-section along profile A–A′, which traverses the Snake River Plain and the Yellowstone volcanic system. The location of the A–A′ profile is marked on the 120 km depth slice from this work. For comparison, we also present the slices at the same depth and along the same profile from SRPY-MT by Kelbert et al. (2012) and CONUS-MT-2023 by Murphy et al. (2023) for the same area. The CONUS-MT-2023 and SRPY-MT models are publicly available via the IRIS Data Management Center's EarthScope Earth Model Collaboration (https://ds.iris.edu/ds/products-/emc/), with contributions from Trabant et al. (2012) and Hutko et al. (2017). The structure of CONUS-MT-2023 in this study area is primarily compiled based on the model obtained by Meqbel et al. (2014).

Despite differences in inversion algorithms, datasets, and mesh, the comparison demonstrates that our model captures structural features broadly consistent with previous studies in terms of large-scale electrical resistivity variations, particularly showing a strong similarity to the CONUS-MT-2023 results. Overall, all models reveal a transition from a moderately resistive upper and middle crust to a lower crust and uppermost mantle characterized by widespread high conductors. These conductors gradually diminish with increasing depth, with the overall resistivity increasing toward approximately 80 km depth.

Notably, SRPY-MT-DABIC was obtained without the need for multiple adjustments of

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regularization parameters, or initial model trials, as required in Kelbert et al. (2012) and Meqbel et al. (2014). This underscores the objectivity and robustness of this D-DABIC inversion, enabling a more automated process and minimizing the influence of subjective intervention, while still ensuring a solution that is similar to their preferred model after multiple trials.

In addition to the large-scale consistencies, notable differences are observed in the depth of the peaks of the widespread high conductors within the lower crust and uppermost mantle. Based on both horizontal and vertical slices, the SRPY-MT model places the center of these features at approximately 50–60 km depth, which is deeper than the ~30–45 km depth observed in both the SRPY-MT-DABIC model and the CONUS-MT-2023 model. Kelbert et al. (2012) explicitly noted that the depth of the conductive peak in their results could vary by more than 10 km depending on the choice of the initial model. This range of variability may explain the discrepancy in central depth among the models. It is worth emphasizing that, in addition to automatically selecting the regularization parameter that maximizes the marginal likelihood, our method is largely insensitive to the choice of the initial model, as demonstrated in the synthetic experiments shown in Figure 7. In other words, the inversion results remain stable across different initial model configurations. This robustness lends further credibility to the similar depth of the conductive center identified in both the SRPY-MT-DABIC and the CONUS-MT-2023.

Moreover, although all models suggest that the lower crust and uppermost mantle directly beneath the Yellowstone volcanic region may not host a large amount of melt, some localized differences are observed, particularly in the 5–20 km depth range beneath the region. In the vertical profile from SRPY-MT-DABIC, several scattered high-conductivity anomalies are detected in this

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depth interval, especially beneath the northeastern portion of the volcanic system. These features are absent in the SRPY-MT model and are not well resolved in the CONUS-MT-2023 model either. Admittedly, given the long-period nature of the data used in this work and the relatively coarse mesh discretization, we cannot assert with certainty the accuracy of these localized features. However, we note that Bennington et al. (2025) recently reported similar isolated high-conductivity zones in the northeastern part of the Yellowstone volcanic region using higher-frequency and more densely distributed MT stations (e.g., Figures 3c and 3d in Bennington et al., 2025). These features were interpreted as pockets of rhyolitic or mixed rhyolitic–basaltic magma in the middle and upper crust. The resemblance of these features between features shown in SRPY-MT-DABIC and those of Bennington et al. (2025) provides additional support for the localized high-conductivity features observed in our model. In contrast, the absence of such anomalies in the SRPY-MT model may reflect the influence of subjective choices in regularization parameters and initial model settings when utilizing the ModEM framework, as noted by Kelbert et al. (2012).

CONCLUSIONS

In this study, we developed a 3-D MT inversion method based on Akaike's Bayesian Information Criterion (ABIC), which statistically determines the optimal balance between prior constraints and data fitting. By maximizing the marginal likelihood function in Bayesian statistics, this method provides an objective criterion for regularization parameter selection that can effectively balance data fitting and model constraints. The original ABIC formulation, however, requires matrix factorization with the target matrix in M dimensions, a requirement that leads to

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high computational costs and makes its direct application in large-scale inversions like 3-D MT problems computationally expensive and impractical.

To mitigate this computational bottleneck, we introduced a low-rank transformation similar to the one used in model-to-data space inversions. This transformation results in a data-space variant of ABIC, reducing the computational complexity by allowing the ABIC indicator to be evaluated in data space. Furthermore, the dense target matrix for the determinant calculation after this transformation is identical to the matrix obtained by the low-rank update inversion in the data space. This implies that the matrix decomposition used for matrix inversion can be directly utilized in the determinant calculation. As a result, applying the data-space variant of ABIC proposed in this paper introduces almost no additional computational burden in data-space inversion; we refer to this approach as the D-DABIC inversion.

Using synthetic MT data, we compared D-DABIC with data-space inversion based on the original ABIC formulation (D-MABIC) and other inversions, including the data-space inversion with empirical L-curve methods, D-Lcurve, and data-space inversion using a cooling scheme, D-Cooling. Each of these methods shows distinct characteristics and trade-offs.

The D-DABIC and D-MABIC share the same statistical framework, but D-MABIC evaluates the likelihood in model space, while D-DABIC operates in data space. In the discussed synthetic experiments, the computational time for D-MABIC is approximately 70 times longer than that for D-DABIC. Based on this comparison, we conclude that the data-space variant of ABIC proposed in this study facilitates applying this statistically rigorous framework to 3-D MT inversions, offering substantial computational gains in cases where the number of model parameters exceeds

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the number of data points. At the same time, it is important to remember this assumption in practical applications. While in most 3-D MT inversions the number of data points is smaller than that of model parameters, we cannot entirely rule out cases where the latter is less than the former. In such scenarios, the original model-space formulation of ABIC would remain the relatively optimal choice.

The D-Lcurve and D-Cooling inversions strongly depend on user-defined parameters. In the synthetic example, the D-Lcurve inversion outcome is sensitive to the range and values of the regularization parameter used to construct the L-curve. The optimal parameter values must be determined empirically, often requiring multiple trials and subjective decisions by users. In addition, although the D-Lcurve method with the empirically determined optimal parameter can recover the main features of the true model, it suffers from instability during the inversion process, with large model modifications occurring even when data misfit improvements are negligible. This instability arises from the regularization strategy, which fails to effectively stabilize the inversion and mitigate the ill-posedness of the problem.

Similarly, the results of D-Cooling depend on multiple factors, including the initial regularization parameter, the decreasing rate, and additional settings such as the misfit reduction threshold. However, unlike D-Lcurve, D-Cooling lacks even a qualitative guideline or criterion for selecting these parameters, making the inversion process heavily reliant on trial and error. The strong dependence of both D-Lcurve and D-Cooling on user-defined parameters poses a significant challenge, particularly in field data applications where no true model is available for reference.

We apply D-DABIC to the Yellowstone–Snake River Plain region using EarthScope USArray

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MT data to showcase its practical applicability. The consistency between the large-scale features shown in our model and previous studies strengthens confidence in the derived electrical structure and provides additional validation of the proposed method's effectiveness. A key advantage of our approach is its objectivity and robustness, as the inversion process does not require any parameter tuning by users. While some discrepancies exist between our model and those of prior studies, such differences are expected due to inherent differences in inversion frameworks and settings, such as variations in inverted data, regularization strategies, and model parameterization. Furthermore, despite the fact that both Meqbel et al. (2014) and Kelbert et al. (2012) employed ModEM, their results also exhibit noticeable variations; however, the overall patterns remain consistent. Therefore, we believe that this general agreement highlights the reliability of our method and its ability to recover robust large-scale conductivity structures. The geological implications of these conductivity features require further analysis, which we discuss in a companion work.

Overall, this work demonstrates the effectiveness of the proposed inversion method and highlights its potential as a powerful tool for 3-D MT investigations. Furthermore, it opens up opportunities for its application in other geophysical inversions, including but not limited to magnetic inversion, gravity inversion, seismic tomography, and joint inversion, where this Bayesian statistical framework can be smoothly adopted. The proposed D-DABIC could serve as a theoretically rigorous yet computationally efficient option for a broader range of inversion topics, as long as the number of data points is smaller than the number of model parameters.

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DATA AVAILABILITY STATEMENT

The D-DABIC inversion method was developed based on the FEMTIC code and is named FEMTIC-DABIC as the corresponding software package. This new package is available at https://github.com/hsong-28/FEMTIC-DABIC, while the original FEMTIC code is freely available at https://github.com/yoshiya-usui/femtic. The USArray MT data utilized in this study are archived with the Incorporated Research Institutions for Seismology (IRIS) and are freely available at http://ds.iris.edu/spud/emtf. The electrical conductivity model of the contiguous United States, CONUS-MT-2023, compiled by Murphy et al. (2023), and the regional 3-D electrical conductivity model of Snake River Plain and Yellowstone region, SRPY-MT, with efforts from Kelbert et al. (2012), are freely accessible at IRIS EMC - Earth Models (http://ds.iris.edu/ds/products/emc-earthmodels/), with contributions from Trabant et al. (2012) and Hutko et al. (2017). Part of the figures are prepared using Generic Mapping Tools (GMT) version 6 (Wessel et al., 2019a, 2019b).

DECLARATION OF COMPETING INTERESTS

The authors declare that they have no competing financial interests or personal relationships that could have appeared to affect the work reported in this paper.

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REFERENCES

Anderson, E., Z. Bai, C. Bischof, L. S. Blackford, J. Demmel, J. Dongarra, J. DuCroz, A. Greenbaum, S. Hammarling, A. McKenney, and D. Sorensen, 2000, LAPACK User's Guide, 3rd ed.: SIAM.

Akaike, H., 1980, Likelihood and the Bayes procedure: Trabajos de Estadística y de Investigación Operativa, **31**, 143–166, https://doi.org/10.1007/BF02888350.

Ansari, S., and J. Craven, 2023, A fully finite‐element based model‐space algorithm for three‐dimensional inversion of magnetotelluric data, Geophysical Journal International, **233**(2), 1245‐1270, https://doi.org/10.1093/gji/ggac519.

Baba, K., 2023, A simple method to evaluate the uncertainty of magnetotelluric forward modeling for practical three-dimensional conductivity structure models, Earth, Planets and Space, **75**(1), 67, https://doi.org/10.1186/s40623-023-01832-5.

Bennington, N., A. Schultz, P. Bedrosian, E. Bowles-Martinez, K. Lynn, M. Stelten, X. Tu, and C. Thurber, 2025, The progression of basaltic–rhyolitic melt storage at Yellowstone Caldera, Nature, **637**(8044), 97–102, https://doi.org/10.1038/s41586-024-08286-z.

Brent, R. P., 1973, Chapters 3–4 in Algorithms for minimization without derivatives: Prentice-Hall.

Constable, S. C., R. L. Parker, and C. G. Constable, 1987, Occam's inversion: A practical algorithm for generating smooth models from electromagnetic sounding data: GEOPHYSICS, **52**, no. 3, 289–300, https://doi.org/10.1190/1.1442303.

Constable, S., A. Orange, and K. Key, 2015, And the geophysicist replied: "Which model do you want?": GEOPHYSICS, **80**, no. 3, E197–E212, https://doi.org/10.1190/geo2014-0381.1.

Chen, H., Z. Ren, and J. Tang, 2025, Magnetotelluric Adaptive Inversion Using Multi‐Resolution Tetrahedral Grids: Application to the North China Craton. Journal of Geophysical Research: Solid Earth, **130**(4), e2024JB030405, https://doi.org/10.1029/2024JB030405.

deGroot‐Hedlin, C., and S. Constable, 1990, Occam's inversion to generate smooth, two-dimensional models from magnetotelluric data: GEOPHYSICS, **55**, no. 12, 1613‐1624, https://doi.org/10.1190/1.1442813.

Dong, S, T. Li, Q. Lü, R. Gao, J. Yang, X. Chen, W. Wei, and Q. Zhou, 2013, Progress in deep lithospheric exploration of the continental China: A review of the SinoProbe: Tectonophysics, **606**, 1–13, https://doi.org/10.1016/j.tecto.2013.05.038.

Dong, H., K. Sun, G. Egbert, A. Kelbert, and N. Meqbel, 2024, Hybrid CPU-GPU solution to regularized divergence-free curl-curl equations for electromagnetic inversion problems, Computers & Geosciences, **184**, 105518, https://doi.org/10.1016/j.cageo.2024.105518.

Diba, D., H. Song, M. Uyeshima, and Y. Usui, 2025, Three-dimensional magnetotelluric inversion with structurally guided regularization constraint: Physics of the Earth and Planetary Interiors, **361**, 107333, https://doi.org/10.1016/j.pepi.2025.107333.

Eftekhari, A., D. Pasadakis, M. Bollhöfer, S. Scheidegger, and O. Schenk, 2021, Block-enhanced precision matrix estimation for large-scale datasets: Journal of Computational Science, **53**,

This paper presented here as accepted for publication in Geophysics prior to copyediting and composition.
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101389, https://doi.org/10.1016/j.jocs.2021.101389.

Egbert, G. D., 2012, Hybrid conjugate gradient Occam algorithms for inversion of multifrequency and multitransmitter EM data: Geophysical Journal International, **190**, 255–266, https://doi.org/10.1111/j.1365-246X.2012.05523.x.

Egbert, G. D., and A. Kelbert, 2012, Computational recipes for electromagnetic inverse problems: Geophysical Journal International, **189**, 251–267, https://doi.org/10.1111/j.1365-246X.2011.05347.x.

Farquharson, C. G., and D. W. Oldenburg, 2004, A comparison of automatic techniques for estimating the regularization parameter in non-linear inverse problems: Geophysical Journal International, **156**, 411–425, https://doi.org/10.1111/j.1365-246X.2004.02190.x.

Finn, C. A., P. A. Bedrosian, W. S. Holbrook, E. Auken, B. R. Bloss, and J. Crosbie, 2022, Geophysical imaging of the Yellowstone hydrothermal plumbing system, Nature, **603**(7902), 643–647, https://doi.org/10.1038/s41586-021-04379-1.

Fukahata, Y., A. Nishitani, and M. Matsu'ura, 2004, Geodetic data inversion using ABIC to estimate slip history during one earthquake cycle with viscoelastic slip-response functions: Geophysical Journal International, **156**, 140–153, https://doi.org/10.1111/j.1365-246X.2004.02122.x.

Fukahata, Y., Y. Yagi, and M. Matsu'ura, 2003, Waveform inversion for seismic source processes using ABIC with two sorts of prior constraints: Comparison between proper and improper formulations: Geophysical Research Letters, **30**, 6, https://doi.org/10.1029/2002GL016293.

Gaedke-Merzhäuser, L., J. van Niekerk, O. Schenk, and R. Haavard, 2023, Parallelized integrated nested Laplace approximations for fast Bayesian inference: Statistics and Computing, **33**, 25–50, https://doi.org/10.1007/s11222-022-10192-1.

Golub, G. H., and C. F. Van Loan, 2013, Matrix computations (Fourth edition): The Johns Hopkins University Press.

Honsho, C., T. Ura, and K. Tamaki, 2012, The inversion of deep-sea magnetic anomalies using Akaike's Bayesian information criterion: Journal of Geophysical Research, **117**, B01105, https://doi.org/10.1029/2011JB008611.

Hutko, A. R., M. Bahavar, C. Trabant, R. T. Weekly, M. V. Fossen, and T. Ahern, 2017, Data Products at the IRIS-DMC: Growth and Usage: Seismological Research Letters, **88**, 892–903, https://doi.org/10.1785/0220160190.

Jackson, D. D., and M. Matsu'ura, 1985, A Bayesian approach to nonlinear inversion: Journal of Geophysical Research: Solid Earth, **90**, 581–591, https://doi.org/10.1029/JB090iB01-p00581.

Kelbert, A., S. Erofeeva, C. Trabant, R. Karstens, and M. Van Fossen, 2018, Taking Magnetotelluric Data out of the Drawer: Eos, **99**, https://doi.org/10.1029/2018EO112859.

Kelbert, A., G. D. Egbert, and C. deGroot-Hedlin, 2012, Crust and upper mantle electrical conductivity beneath the Yellowstone Hotspot Track: Geology, 40, 447–450, https://doi.org/10.1130/G32655.1.

Kelbert, A., N. Meqbel, G. D. Egbert, and K. Tandon, 2014, ModEM: A modular system for inversion of electromagnetic geophysical data: Computers & Geosciences, **66**, 40–53, https://doi.org/10.1016/j.cageo.2014.01.010.

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Kordy, M., P. Wannamaker, V. Maris, E. Cherkaev, and G. Hill, 2016, Three-dimensional magnetotelluric inversion including topography using deformed hexahedral edge finite elements and direct solvers parallelized on symmetric multiprocessor computers – Part II: Direct data-space inverse solution: Geophysical Journal International, **204**, 94–110, https://doi.org/10.1093/gji/ggv411.

Li, Y., and D. W. Oldenburg, 1999, 3 - D inversion of DC resistivity data using an L - curve criterion: SEG Technical Program Expanded Abstracts 1999, 251 - 254, https://doi.org/10.1190/1.1820992.

Llorente, F., L. Martino, D. Delgado, and J. Lopez-Santiago, 2023, Marginal likelihood computation for model selection and hypothesis testing: An extensive review: SIAM Review, **65**, 3–58, https://doi.org/10.1137/20M1310849.

Meqbel, N., G. D. Egbert, P. E. Wannamaker, A. Kelbert, and A. Schultz, 2014, Deep electrical resistivity structure of the northwestern U.S. derived from 3-D inversion of USArray magnetotelluric data: Earth and Planetary Science Letters, **402**, 290–304, https://doi.org/10.1016/j.epsl.2013.12.026.

Miensopust, M. P., 2017, Application of 3-D Electromagnetic Inversion in Practice: Challenges, Pitfalls and Solution Approaches: Surveys in Geophysics, **38**, 869–933, https://doi.org/10.1007/s10712-017-9435-1.

Mitsuhata, Y., Uchida, T., and H. Amano, 2002, 2.5-D inversion of frequency-domain electromagnetic data generated by a grounded-wire source: GEOPHYSICS, **67**, 1753–1768, https://doi.org/10.1190/1.1527076.

Mitsuhata, Y., and T. Uchida, 2002, Chapter 9: 2-D inversion of frequency-domain EM data caused by a 3-D source: In Methods in Geochemistry and Geophysics, **35**, 153–171, https://doi.org/10.1016/S0076-6895(02)80091-9.

Mitsuhata, Y., 2004, Adjustment of regularization in ill-posed linear inverse problems by the empirical Bayes approach: Geophysical Prospecting, **52**(3), 213–239, https://doi.org/10.1111/j.1365-2478.2004.00412.x.

Munch, F. D., and A. Grayver, 2023, Multi-scale imaging of 3-D electrical conductivity structure under the contiguous US constrains lateral variations in the upper mantle water content, Earth and Planetary Science Letters, **602**, 117939, https://doi.org/10.1016/j.epsl.2022.117939.

Murata, Y., 1993, Estimation of optimum average surficial density from gravity data: An objective Bayesian Approach: Journal of Geophysical Research: Solid Earth, **98**, 12097–12109, https://doi.org/10.1029/93JB00192.

Murphy, B. S., P. Bedrosian, and A. Kelbert, 2023, Geoelectric constraints on the Precambrian assembly and architecture of southern Laurentia, in Laurentia: Turning Points in the Evolution of a Continent, S. J. Whitmeyer, M. L. Williams, D. A. Kellett, and B. Tikoff, Geological Society of America.

Ogawa, Y., and T. Uchida, 1996, A two-dimensional magnetotelluric inversion assuming Gaussian static shift: Geophysical Journal International, **126**, 69–76, https://doi.org/10.1111/j.1365-246X.1996.tb05267.x.

Pasadakis, D., M. Bollhöfer, and O. Schenk, 2023, Sparse Quadratic Approximation for Graph

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Learning: IEEE Transactions on Pattern Analysis and Machine Intelligence, **45**, .11256-11269, https://doi.org/10.1109/TPAMI.2023.3263969.

Rasmussen, C. E., and C. K. I. Williams, 2006, Gaussian processes for machine learning: MIT Press.

Ren, Z., and T. Kalscheuer, 2020, Uncertainty and Resolution Analysis of 2D and 3D Inversion Models Computed from Geophysical Electromagnetic Data: Surveys in Geophysics, **41**, 47–112, https://doi.org/10.1007/s10712-019-09567-3.

Rodi, W., and R. L. Mackie, 2001, Nonlinear conjugate gradients algorithm for 2-D magnetotelluric inversion: GEOPHYSICS, **66**, 174–187, https://doi.org/10.1190/1.1444893.

Schultz, A., G. D. Egbert, A. Kelbert, T. Peery, V. Clote, B. Fry, S. Erofeeva, and staff of the National Geoelectromagnetic Facility and their contractors (2006–2018), 2025, USArray TA Magnetotelluric Transfer Functions, https://doi.org/10.17611/DP/EMTF/USARRAY/TA.

Shimizu, H., A. Yoneda, K. Baba, H. Utada, and N. A. Palshin, 2011, Sq effect on the electromagnetic response functions in the period range between 10^4 and 10^5 s: Sq field effect on EM responses: Geophysical Journal International, **186**, 193–206, https://doi.org/10.1111/j.1365-246X.2011.05036.x.

Siripunvaraporn, W., and G. Egbert, 2000, An efficient data‐subspace inversion method for 2-D magnetotelluric data: GEOPHYSICS, **65**, 791‐803, https://doi.org/10.1190/1.1444778.

Siripunvaraporn, W., and G. Egbert, 2007, Data space conjugate gradient inversion for 2-D magnetotelluric data: Geophysical Journal International, **170**, 986–994, https://doi.org/10.1111/j.1365-246X.2007.03478.x.

Siripunvaraporn, W., G. Egbert, Y. Lenbury, and M. Uyeshima, 2005, Three-dimensional magnetotelluric inversion: Data-space method: Physics of the Earth and Planetary Interiors, **150**, 3–14, https://doi.org/10.1016/j.pepi.2004.08.023.

Siripunvaraporn, W., and W. Sarakorn, 2011, An efficient data space conjugate gradient Occam's method for three-dimensional magnetotelluric inversion: A hybrid DCG-Occam inversion for 3-D MT data: Geophysical Journal International, **186**, 567–579, https://doi.org/10.1111-/j.1365-246X.2011.05079.x.

Song, H., Y. Usui, T. Koyama, D. Diba, M. Uyeshima, K. Baba, P. Yu, and B. Yang, 2024, The use of L-curve criteria in non-linear inverse problem: In Paper Presented at the 26th EM Induction Workshop, Beppu, Japan.

Song, H., Y. Usui, M. Uyeshima, P. Yu, D. Diba, B. Yang, K. Baba, T. Koyama, (Accepted), Revealing the Non-uniqueness Inherent in MT inversion: A Comparative Study of Influential Factors and Algorithm-Dependent Uncertainties in FEMTIC and ModEM with USArray MT Data: Survey in Geophysics.

Smith, J. T., and J. R. Booker, 1988, Magnetotelluric inversion for minimum structure: GEOPHYSICS, **53**, 1565–1576, https://doi.org/10.1190/1.1442438.

Tikhonov, A. N., and V. Y. Arsenin, 1977, Solution of Ill-Posed Problems, Winton and Sons, Washington DC.

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Thiel, S., B. R. Goleby, M. J. Pawley, and G. Heinson, 2020, AusLAMP 3D MT imaging of an intracontinental deformation zone, Musgrave Province, Central Australia, Earth, Planets and Space, **72**(1), 98, https://doi.org/10.1186/s40623-020-01223-0.

Trabant, C., A. R. Hutko, M. Bahavar, R. Karstens, T. Ahern, and R. Aster, 2012, Data Products at the IRIS DMC: Stepping Stones for Research and Other Applications: Seismological Research Letters, **83**, 846–854, https://doi.org/10.1785/0220120032.

Tsuyuki, T., A. Kobayashi, R. Kai, T. Kimura, and S. Itaba, 2021, Joint inversion of strain and tilt data using the Akaike's Bayesian information criterion to map detailed slip distributions of short-term slow slip events: Earth, Planets and Space, **73**, 181, https://doi.org/10.1186-/s40623-021-01517-x.

Uchida, T., 1993, Smooth 2-D Inversion for Magnetotelluric Data Based on Statistical Criterion ABIC: Journal of Geomagnetism and Geoelectricity, **45**, 841–858, https://doi.org/10.5636-/jgg.45.841.

Uchida, T., and Y. Sasaki, 2006, Stable 3D Inversion of MT Data and its Application to Geothermal Exploration. Exploration Geophysics, **37**(3), 223–230. https://doi.org/10.1071/-EG06223.

Usui, Y., 2015, 3-D inversion of magnetotelluric data using unstructured tetrahedral elements: Applicability to data affected by topography: Geophysical Journal International, **202**, 828–849, https://doi.org/10.1093/gji/ggv186.

Usui, Y., Y. Ogawa, K. Aizawa, W. Kanda, T. Hashimoto, T. Koyama, Y. Yamaya, and T. Kagiyama, 2017, Three-dimensional resistivity structure of Asama Volcano revealed by data-space magnetotelluric inversion using unstructured tetrahedral elements: Geophysical Journal International, **208**, 1359–1372, https://doi.org/10.1093/gji/ggw459.

Usui, Y., M. Uyeshima, H. Hase, H. Ichihara, K. Aizawa, T. Koyama, S. Sakanaka, T. Ogawa, Y. Yamaya, T. Nishitani, R. Asamori, Y. Ogawa, R. Yoshimura, S. Takakura, M. Mishina, and Y. Morita, 2024, Three‐Dimensional Electrical Resistivity Structure Beneath a Strain Concentration Area in the Back‐Arc Side of the Northeastern Japan Arc: Journal of Geophysical Research: Solid Earth, **129**, e2023JB028522, https://doi.org/10.1029-/2023JB028522.

Usui, Y., M. Uyeshima, T. Ogawa, R. Yoshimura, N. Oshiman, S. Yamaguchi, H. Toh, H. Murakami, K. Aizawa, T. Tanbo, Y. Ogawa, T. Nishitani, S. Sakanaka, M. Mishina, H. Satoh, T. Goto, T. Kasaya, T. Mogi, and Y. Yamaya, 2021, Electrical Resistivity Structure Around the Atotsugawa Fault, Central Japan, Revealed by a New 2D Inversion Method Combining Wideband‐MT and Network‐MT Data Sets: Journal of Geophysical Research: Solid Earth, **126**, https://doi.org/10.1029/2020JB020904.

Usui, Y., T. Kasaya, Y. Ogawa, and H. Iwamoto, 2018, Marine magnetotelluric inversion with an unstructured tetrahedral mesh. Geophysical Journal International, **214**(2), 952–974. https:-//doi.org/10.1093/gji/ggy171.

Varılsüha, D., 2024, A novel hybrid finite element – Finite difference approach for 3D Magnetotelluric modeling. Computers & Geosciences, **188**, 105614, https://doi.org/10.1016-/j.cageo.2024.105614.

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Wessel, P., J. F. Luis, L. Uieda, R. Scharroo, F. Wobbe, W. H. F. Smith, and D. Tian, 2019a, The generic mapping tools (version 6) [Software]: Zenodo, https://doi.org/10.5281/zenodo.-3407866.

Wessel, P., J. F. Luis, L. Uieda, R. Scharroo, F. Wobbe, W. H. F. Smith, and D. Tian, 2019b, The Generic Mapping Tools Version 6: Geochemistry, Geophysics, Geosystems, **20**, 5556–5564, https://doi.org/10.1029/2019gc008515.

Yabuki, T., and M. Matsu'ura, 1992, Geodetic data inversion using a Bayesian information criterion for spatial distribution of fault slip: Geophysical Journal International, **109**, 363–375, https://doi.org/10.1111/j.1365-246X.1992.tb00102.x.

Zhang, L., T. Koyama, H. Utada, P. Yu, and J. Wang, 2012, A regularized three-dimensional magnetotelluric inversion with a minimum gradient support constraint: 3-D MT inversion with minimum gradient support: Geophysical Journal International, **189**, 296–316, https:-//doi.org/10.1111/j.1365-246X.2012.05379.x.

Zhdanov, M. S., 2002, Geophysical inverse theory and regularization problems: Elsevier Science, Amsterdam, Boston.

Zhdanov, M. S., 2015, Ill-Posed Problems and the Methods of Their Solution: In Inverse Theory and Applications in Geophysics (pp. 33–61), Elsevier, https://doi.org/10.1016/B978-0-444-62674-5.00002-5.

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Table Caption

Table 1. Computation times for different inversions

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975 **Table 1.** Computation times for different inversions

Method	CPU time per trial (hour)	# of trials required	Total CPU time (hour)
D-DABIC	7.9	1	7.9
D-MABIC	553.6	1	553.6
D-Lcurve	~1 to 4	16	33.8
D-Cooling	~1 to 4	12	26.2

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Figure Captions

Figure 1. Flowchart of 3-D data-space MT inversion based on the data-space variant of ABIC (D-DABIC).

Figure 2. The Vertical Cross-Section (V-CS, top row) and Horizontal Cross-Sections (H-CS, second row) of the simplified synthetic model based on the electrical structure around the Atotsugawa Fault system in central Japan, imaged by Usui et al. (2021). The vertical cross-section is taken along the blue line shown in the H-CSs, while the three H-CSs correspond to depths of 1km, 5 km, and 25 km. Black inverted triangles mark MT stations.

Figure 3. The vertical cross-section (first column) and horizontal cross-sections (second to fourth columns) of the D-DABIC inversion results, arranged from left to right for cases adding Gaussian random noise with standard deviations of 2%, 3%, and 5% of the maximum amplitude of the off-diagonal impedance tensor components.

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the ‘optimal’ model shown in Figure 8d.

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Figures

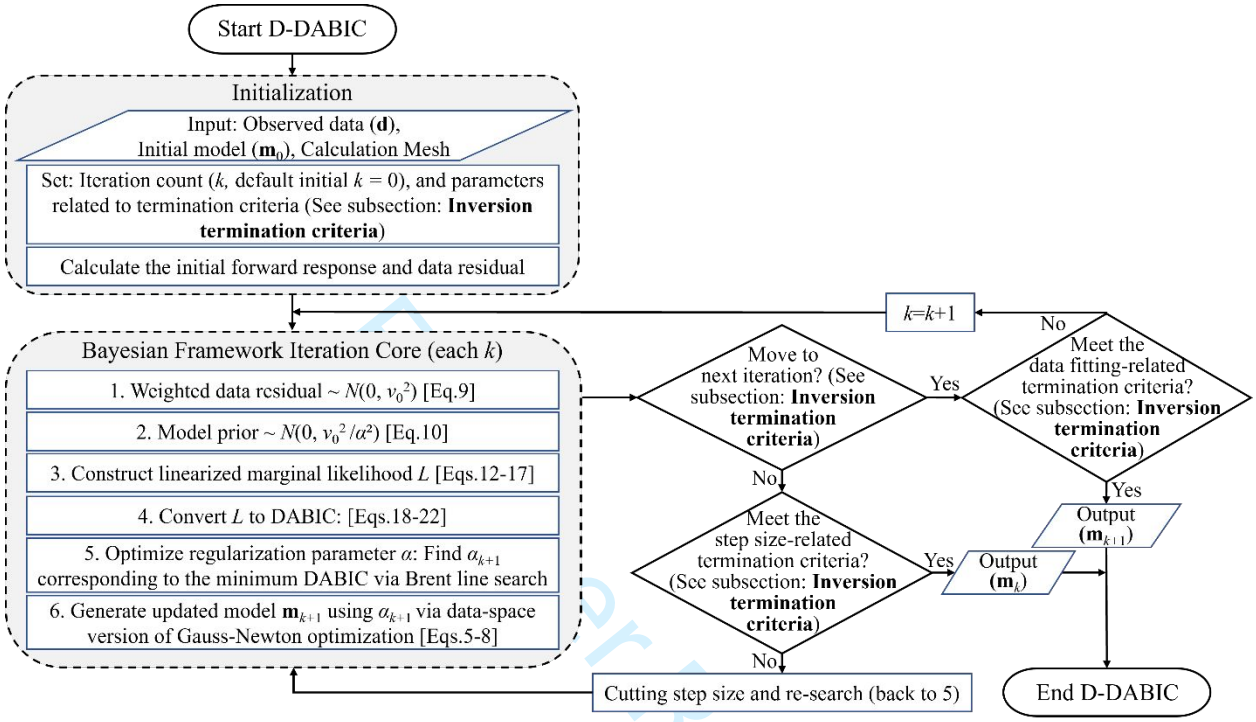


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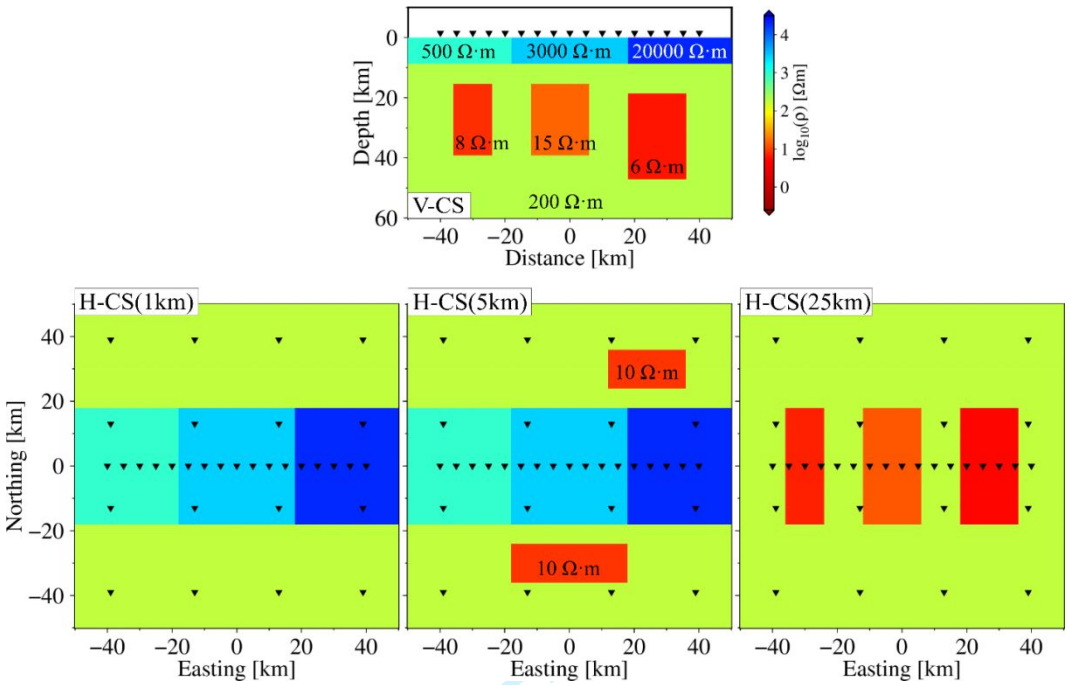


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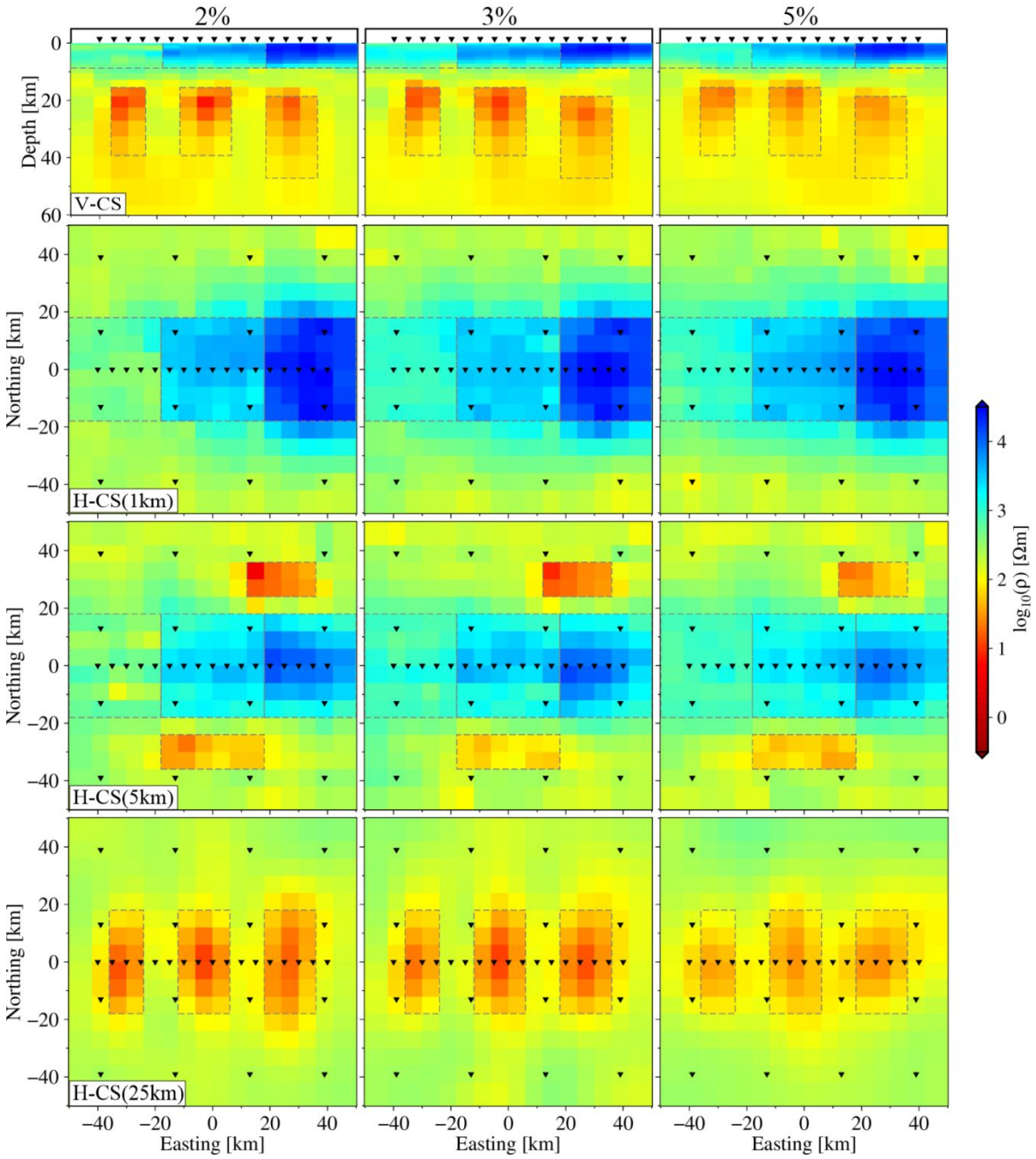


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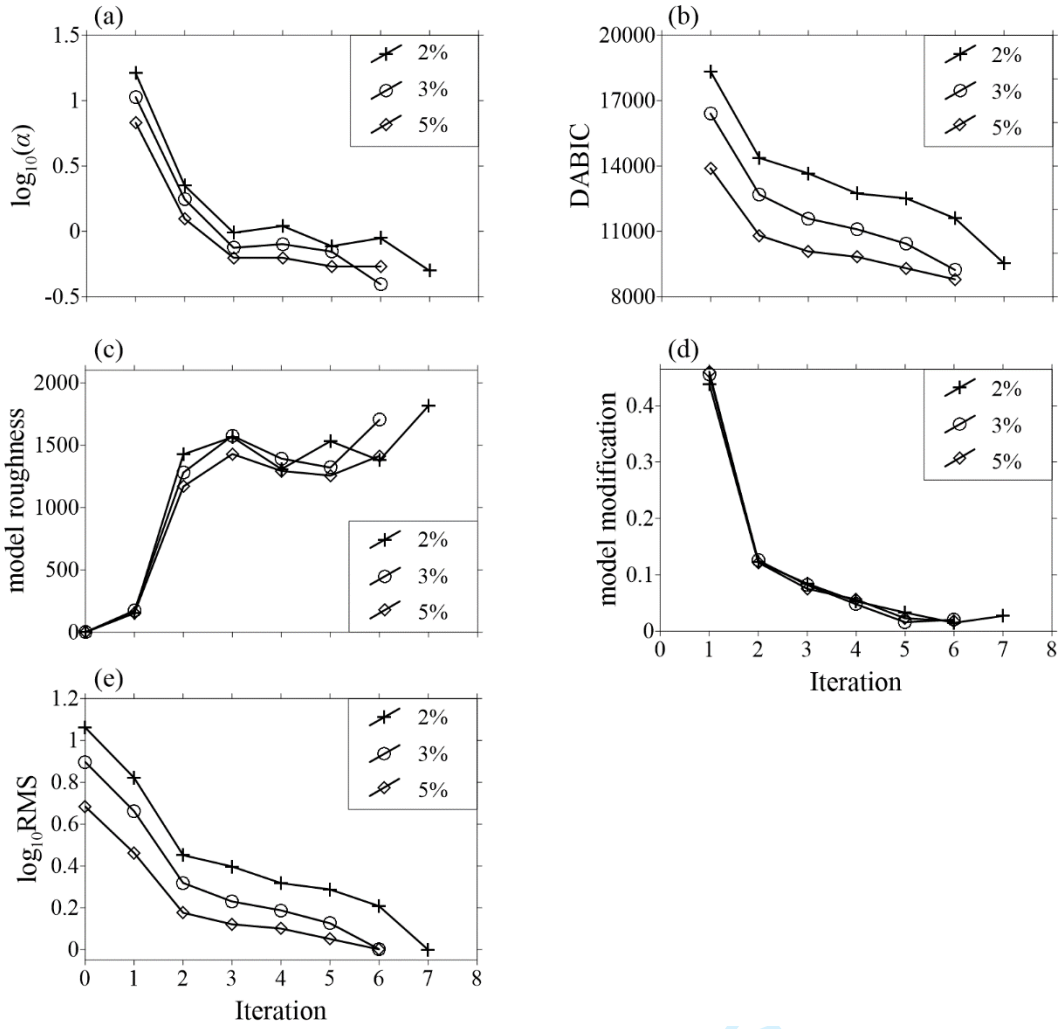


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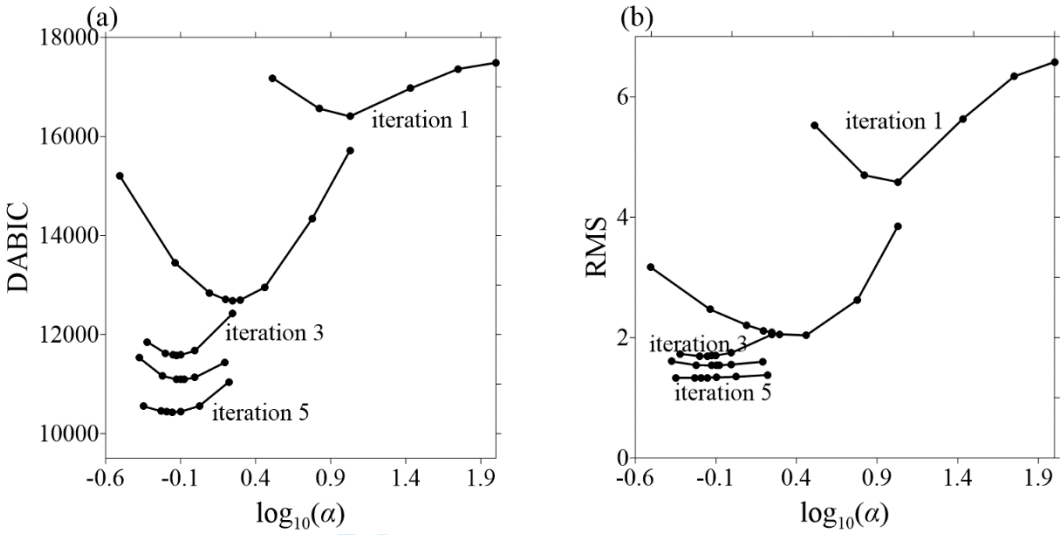


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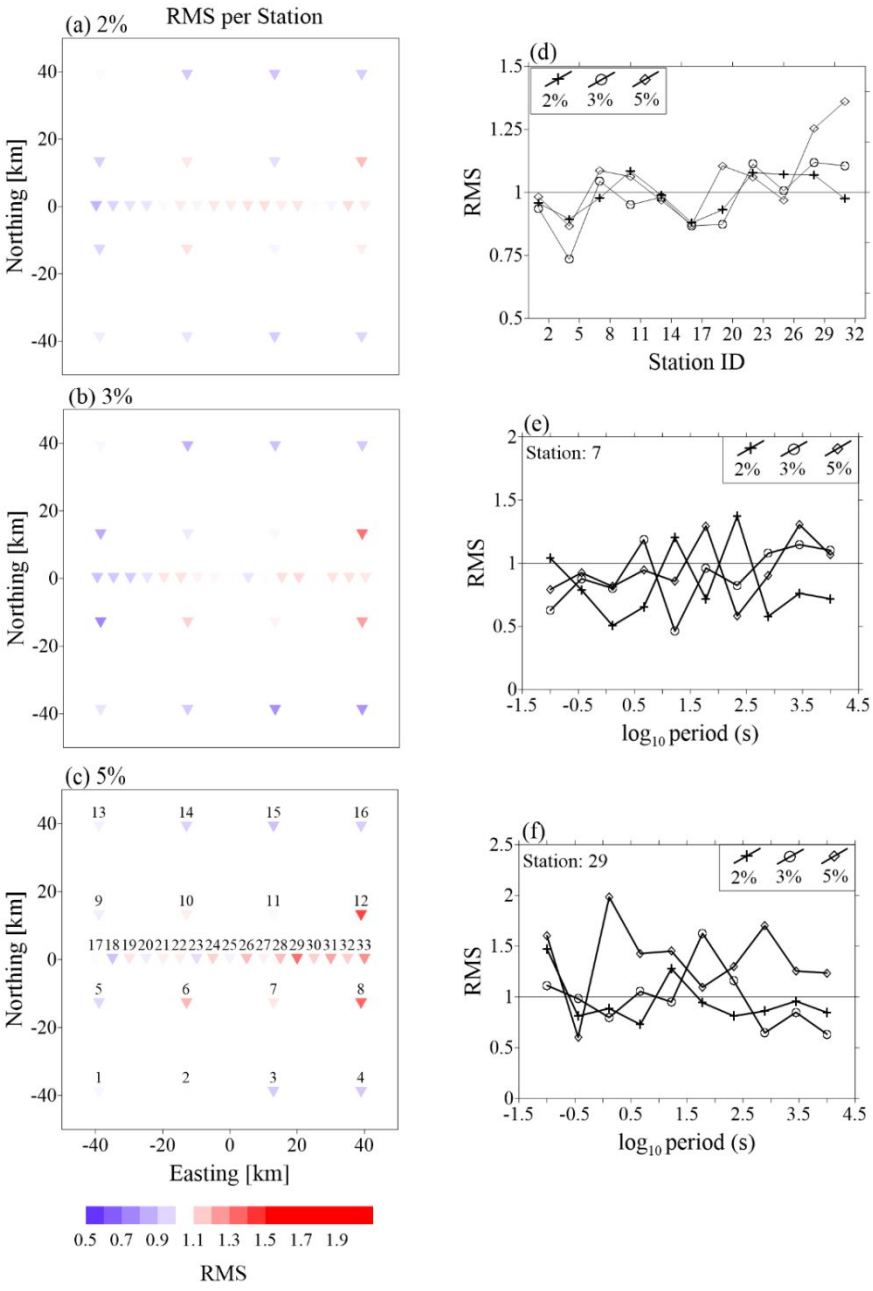


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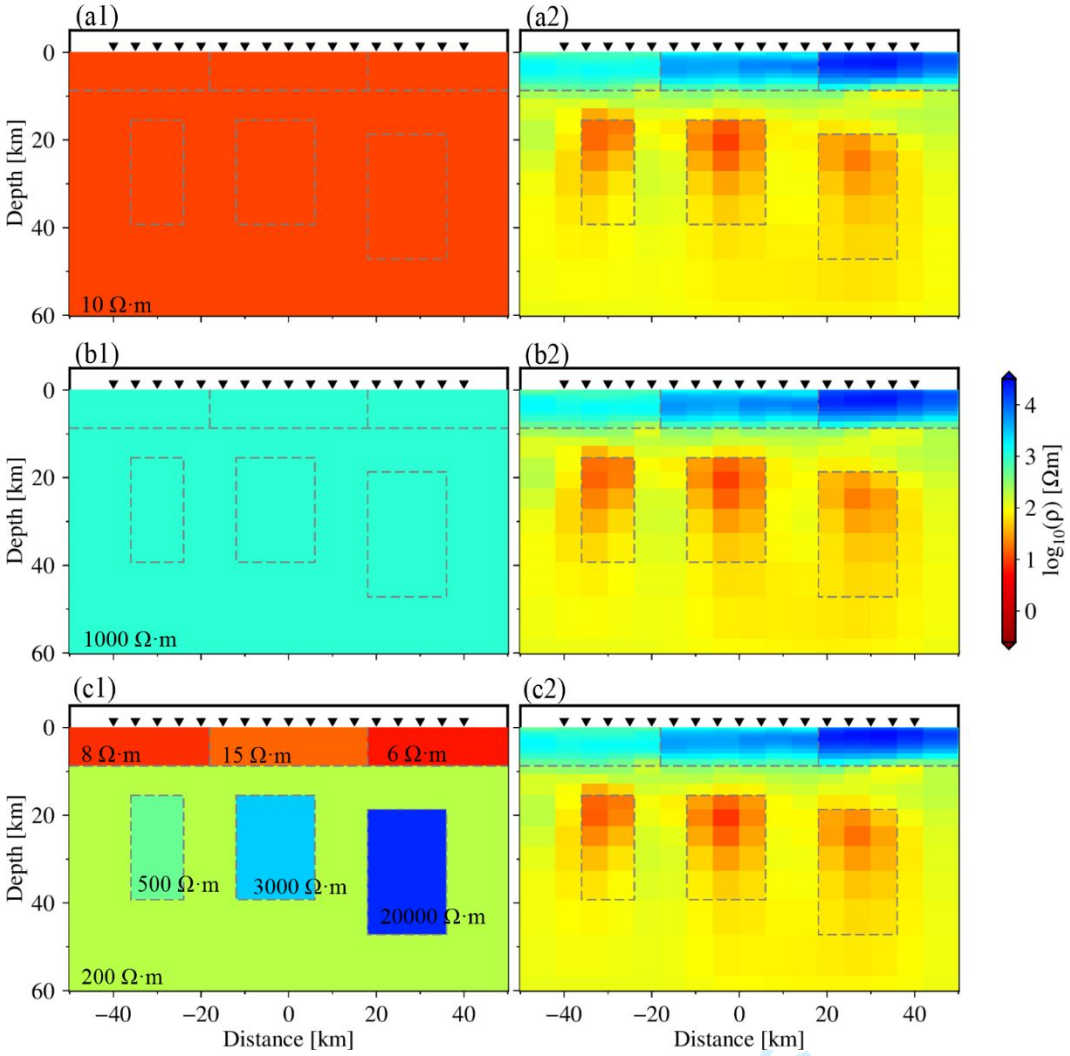


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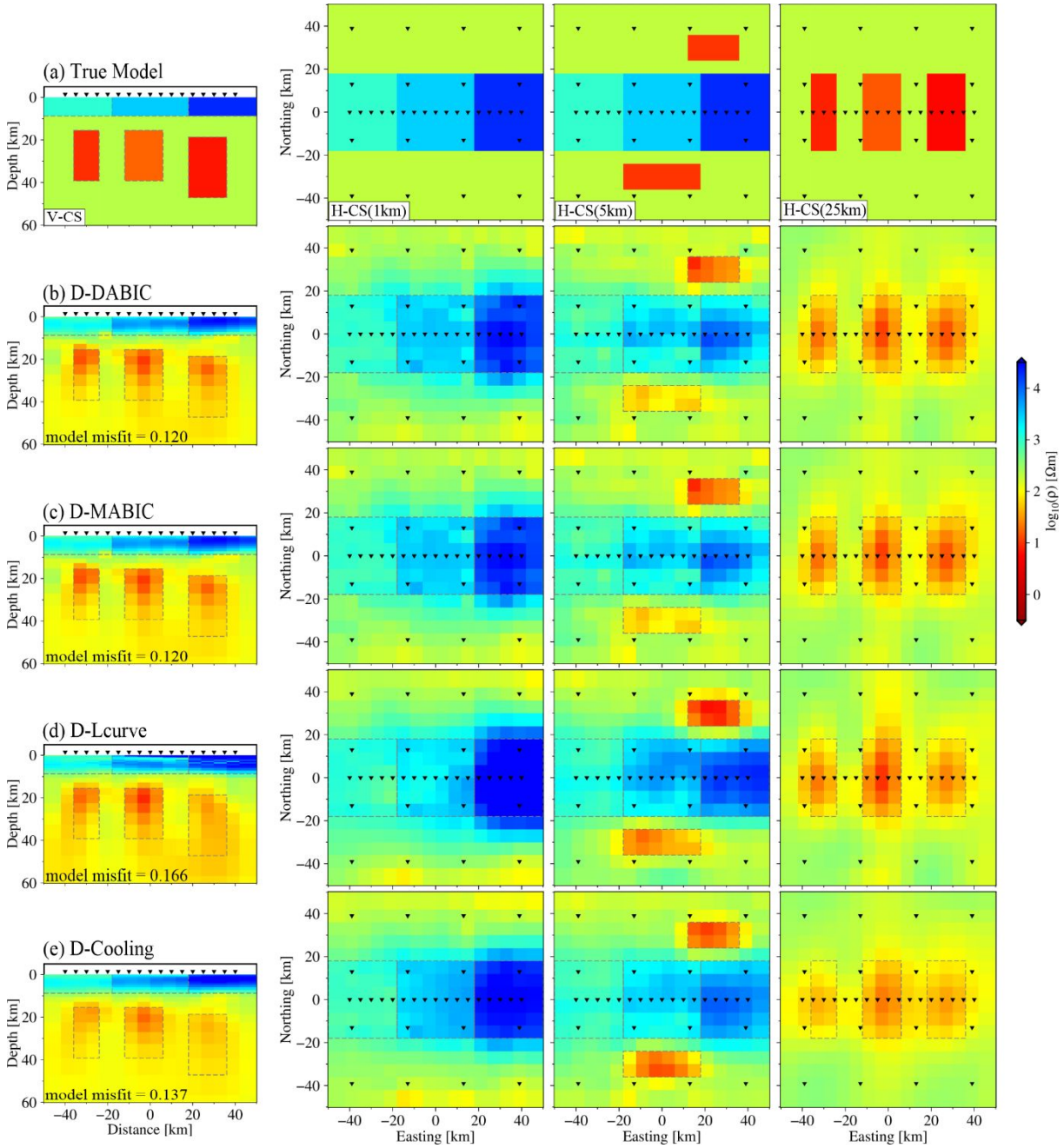


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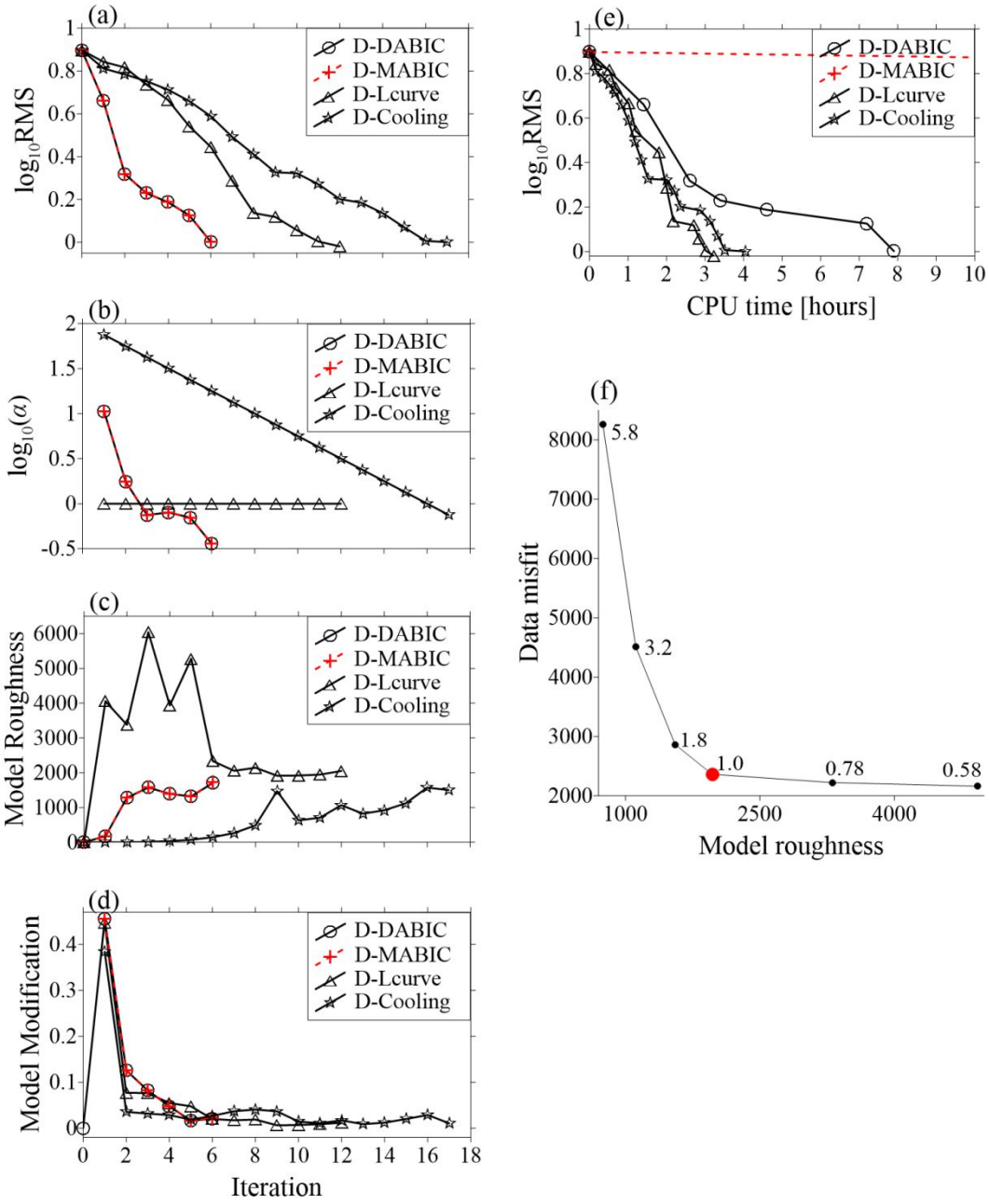


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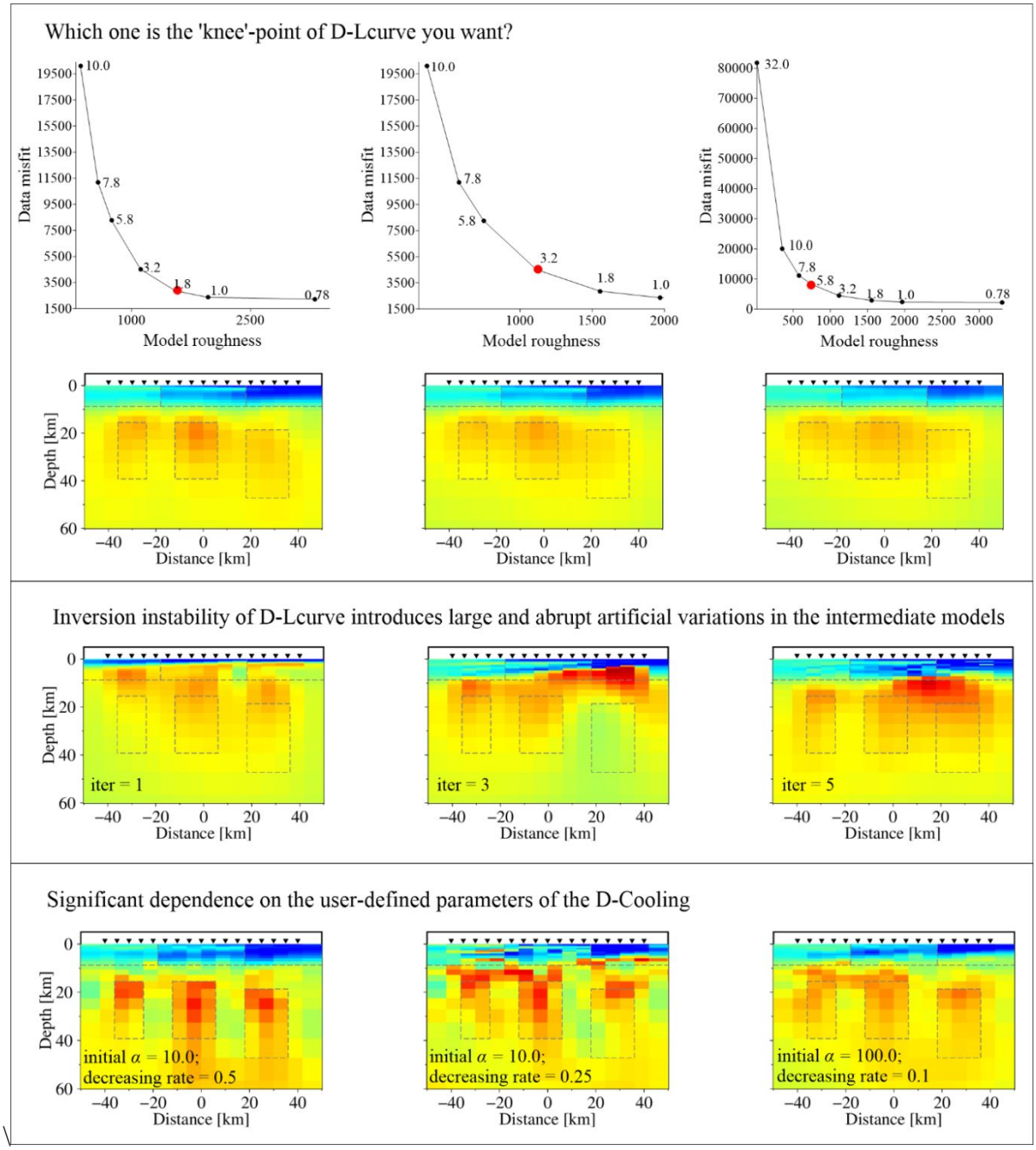


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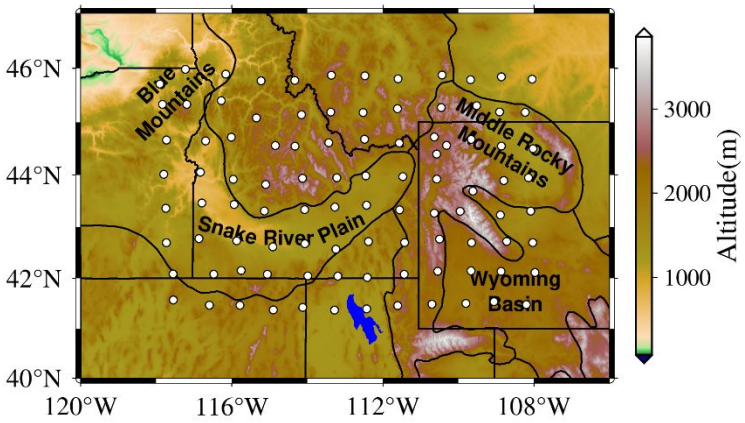


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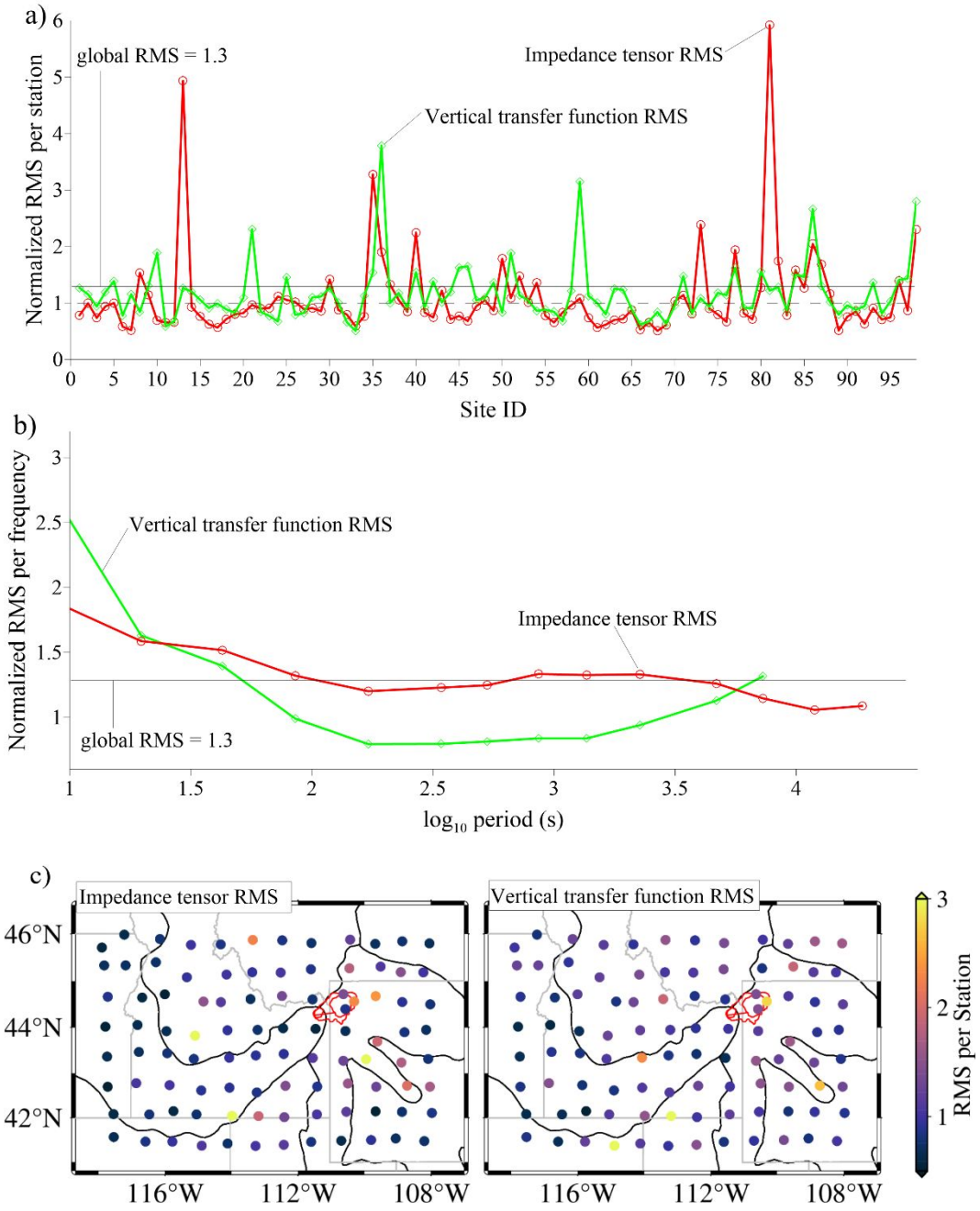


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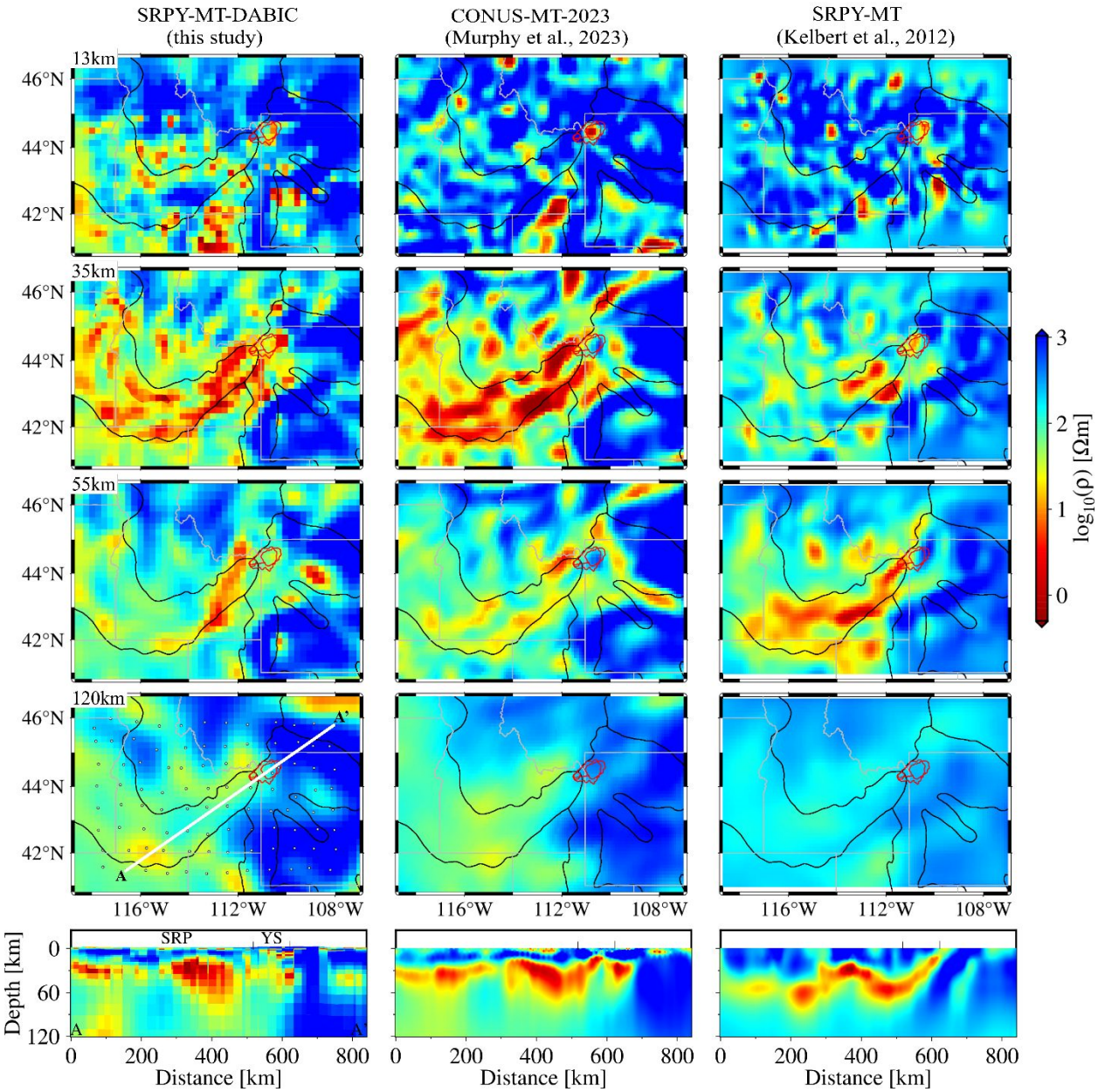


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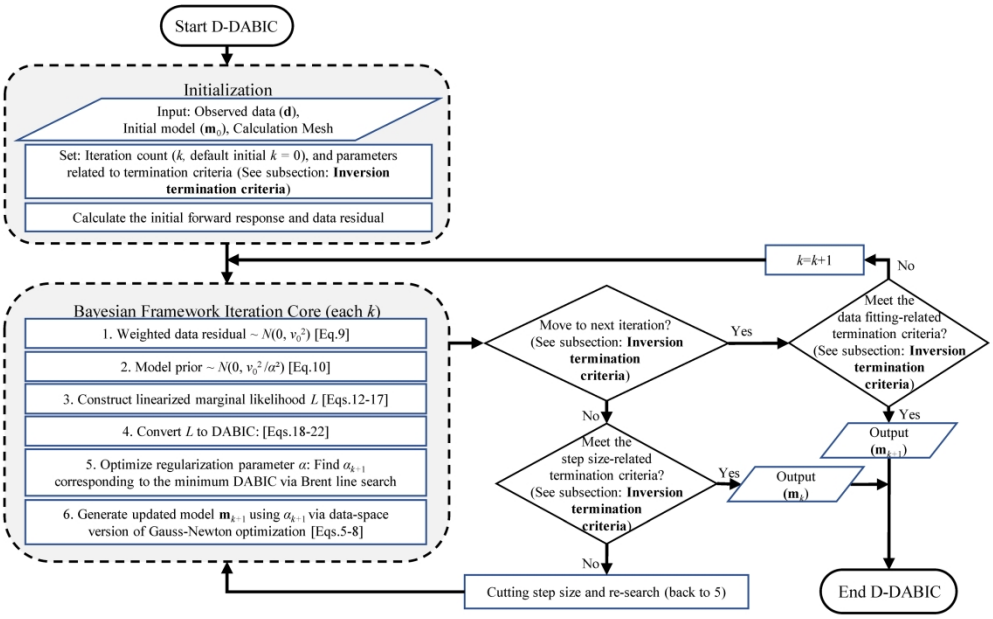


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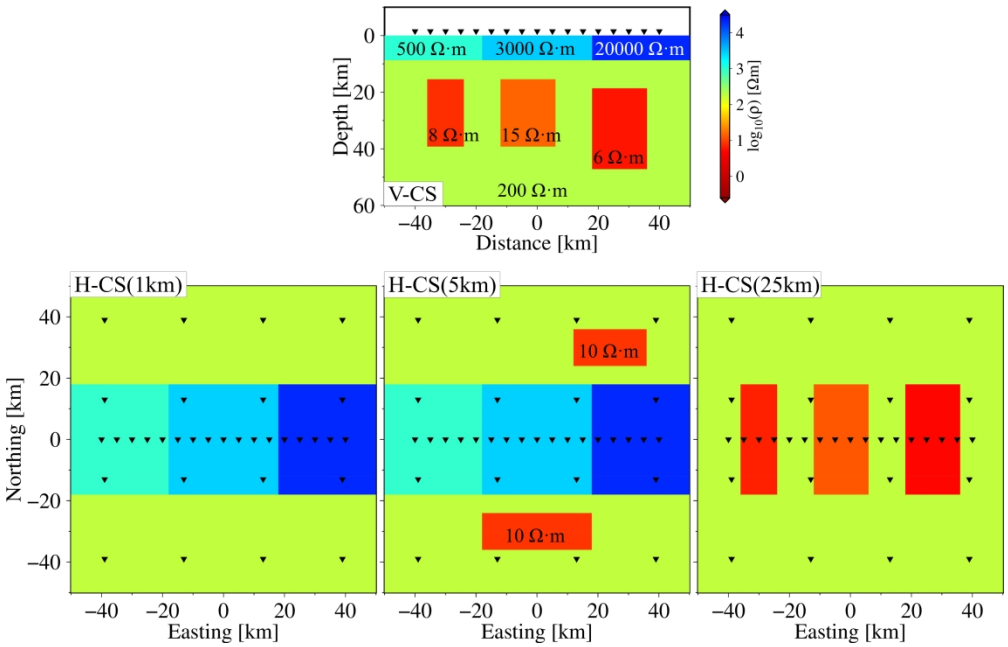


Figure 2. The Vertical Cross-Section (V-CS, top row) and Horizontal Cross-Sections (H-CS, second row) of the simplified synthetic model based on the electrical structure around the Atotsugawa Fault system in central Japan, imaged by Usui et al. (2021). The vertical cross-section is taken along the blue line shown in the H-CSs, while the three H-CSs correspond to depths of 1km, 5 km, and 25 km. Black inverted triangles mark MT stations.

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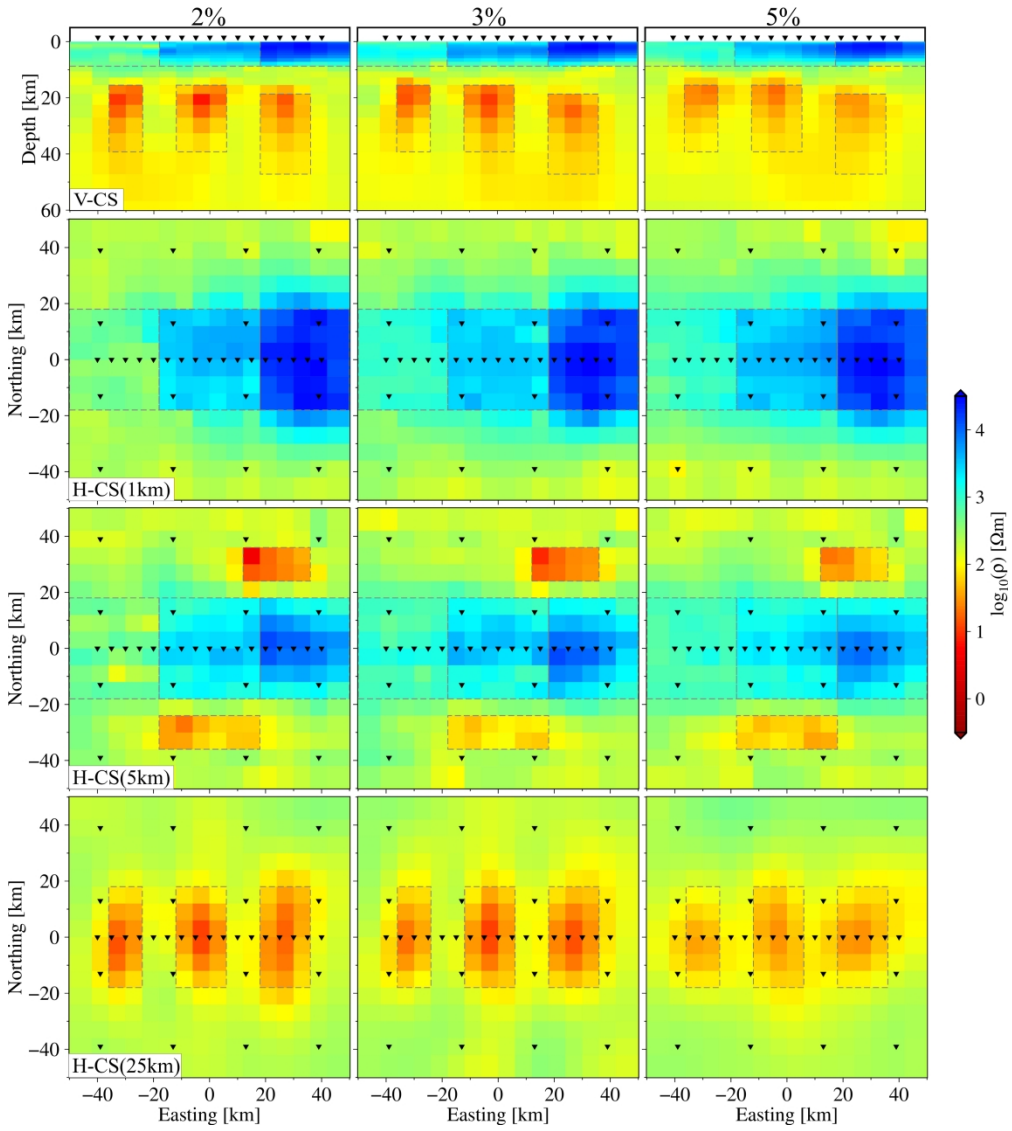


Figure 3. The vertical cross-section (first column) and horizontal cross-sections (second to fourth columns) of the D-DABIC inversion results, arranged from left to right for cases adding Gaussian random noise with standard deviations of 2%, 3%, and 5% of the maximum amplitude of the off-diagonal impedance tensor components.

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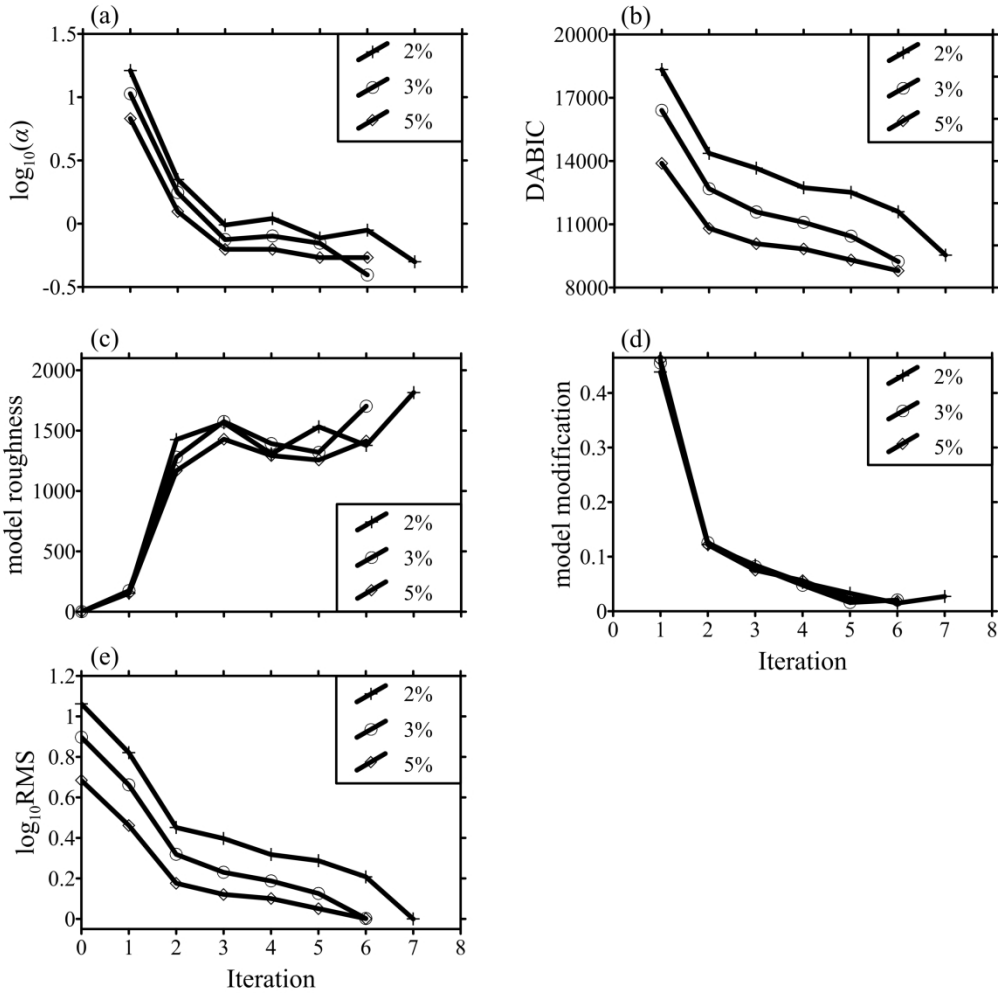


Figure 4. Evolution of inversion parameters as a function of iteration for models obtained using different noisy datasets: (a) regularization parameter on a logarithmic scale ($\log_{10}(\sigma)$), (b) ABIC value, (c) model roughness, (d) absolute mean model modification, and (e) data RMS on a logarithmic scale ($\log_{10}$RMS).

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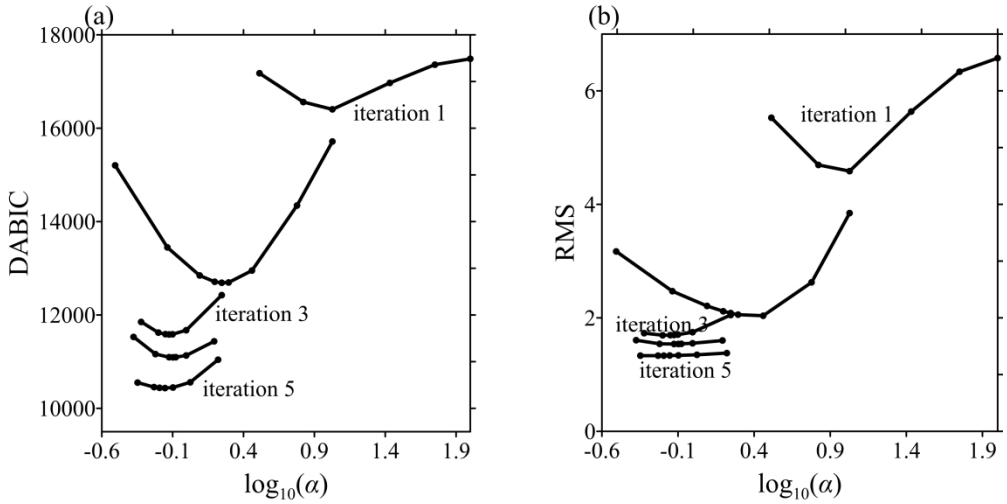


Figure 5. DABIC and data RMS as a function of regularization parameters for the inversion case that contains 3% Gaussian random noise.

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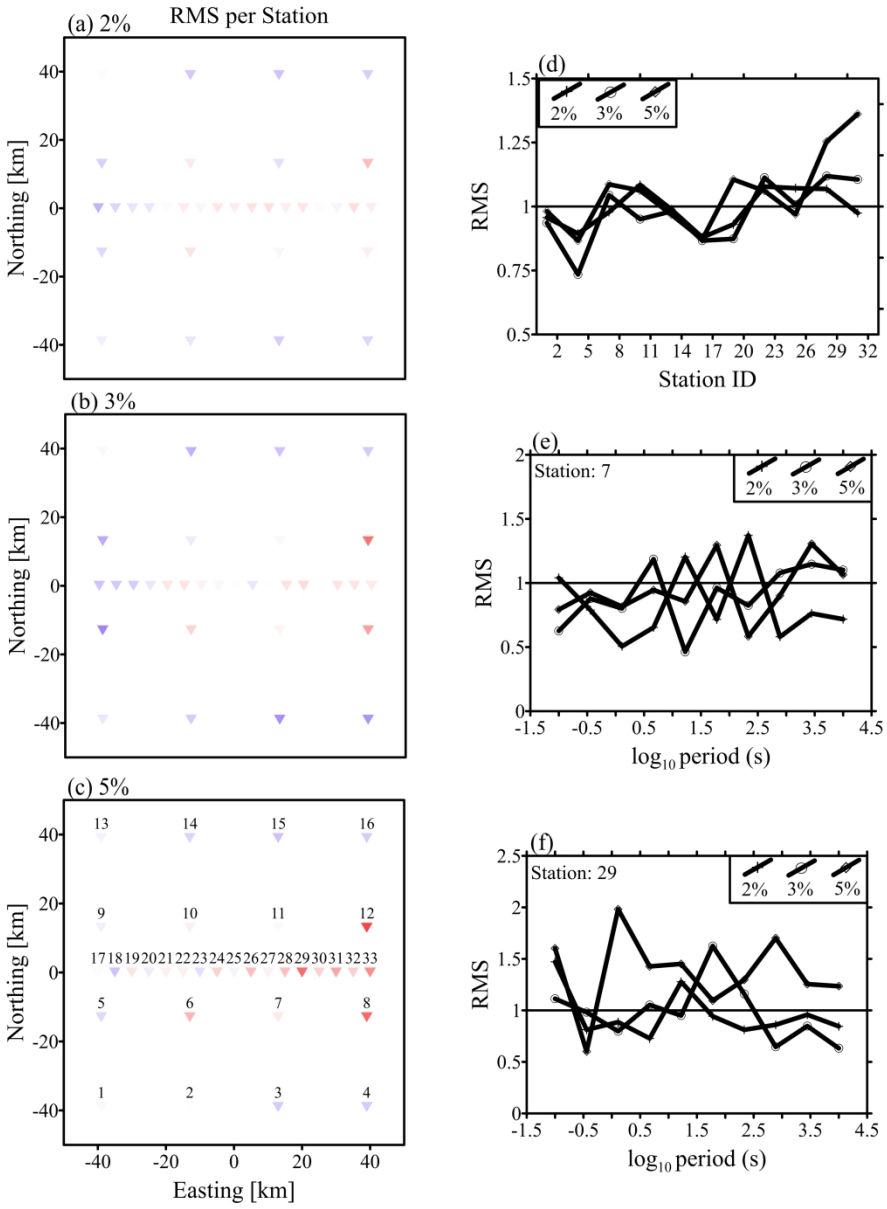


Figure 6. Final RMS misfit across all frequencies for each station under three different noise levels: (a) 2%, (b) 3%, and (c) 5%. The numbers above each station in (c) indicate the station IDs. (d) RMS misfit statistics were computed for every three stations under the three noise levels. (e, f) Relationship between data residuals and frequency for two randomly selected stations: (e) station 7 and (f) station 29.

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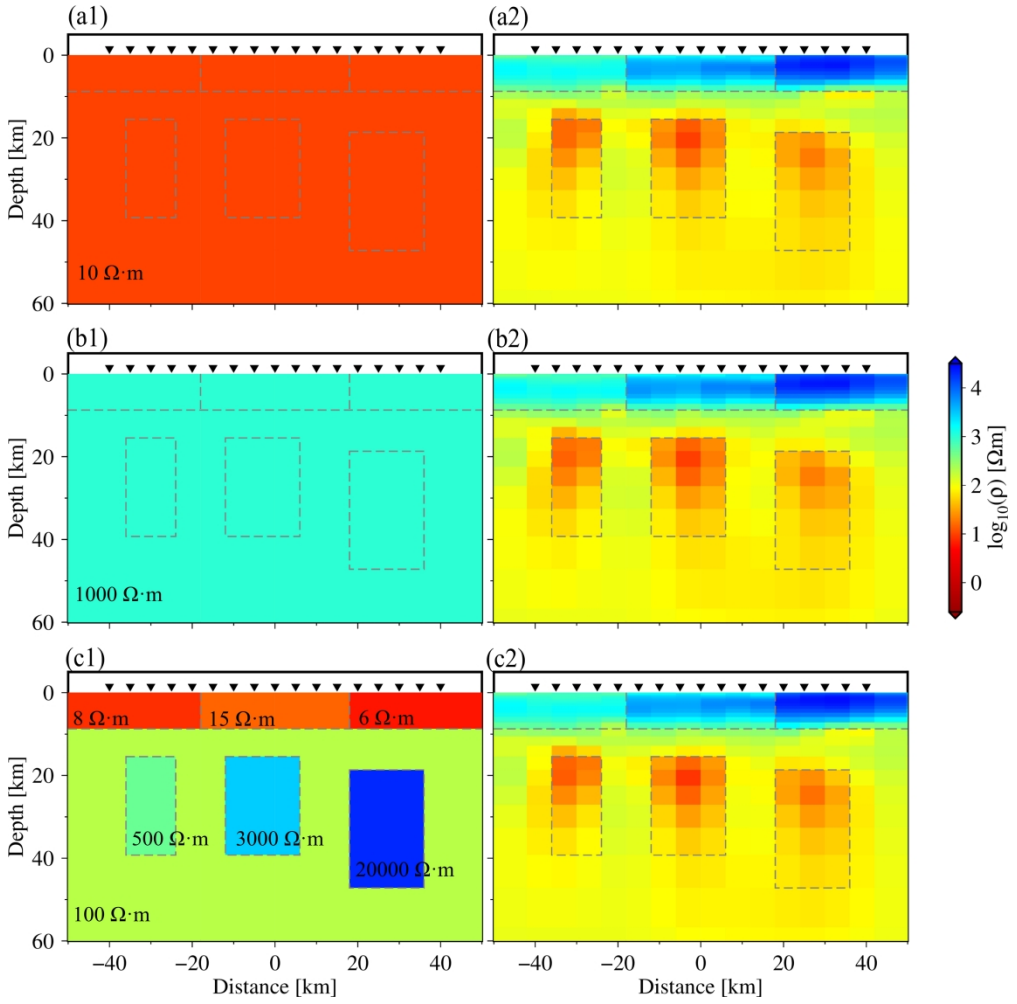


Figure 7. Different initial models: (a1) 10 Ω·m, (b1) 1000 Ω·m, (c1) rough model, and the corresponding final D-DABIC's solutions: (a2), (b2), (c2).

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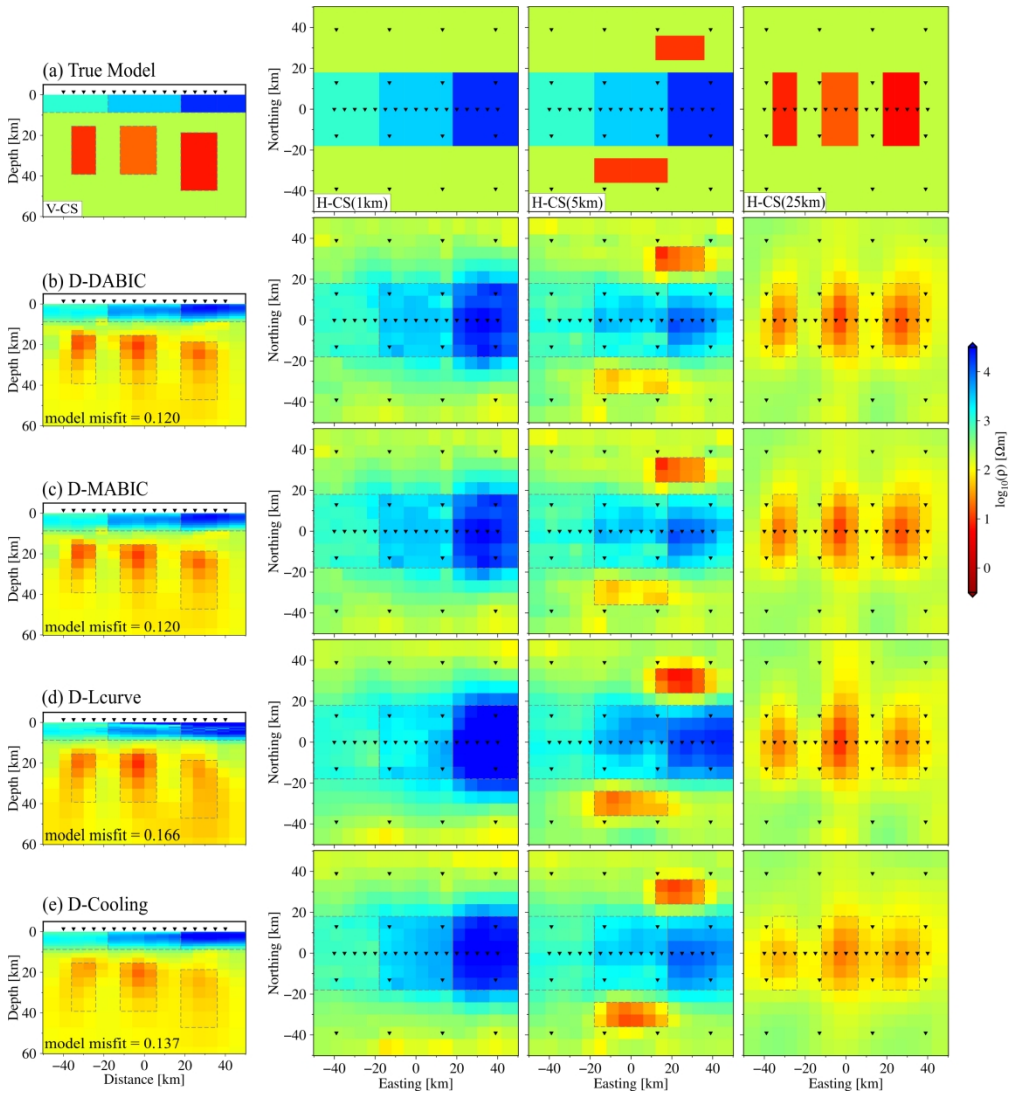


Figure 8. Comparison of inversion results using different methods. (a) True model, and final inversion results obtained using: (b) D-DABIC, (c) D-MABIC, (d) D-Lcurve, and (e) D-Cooling. The slices shown from left to right represent a vertical cross-section along the profile with dense MT stations, and horizontal slices at depths of 1 km, 5 km, and 25 km, respectively.

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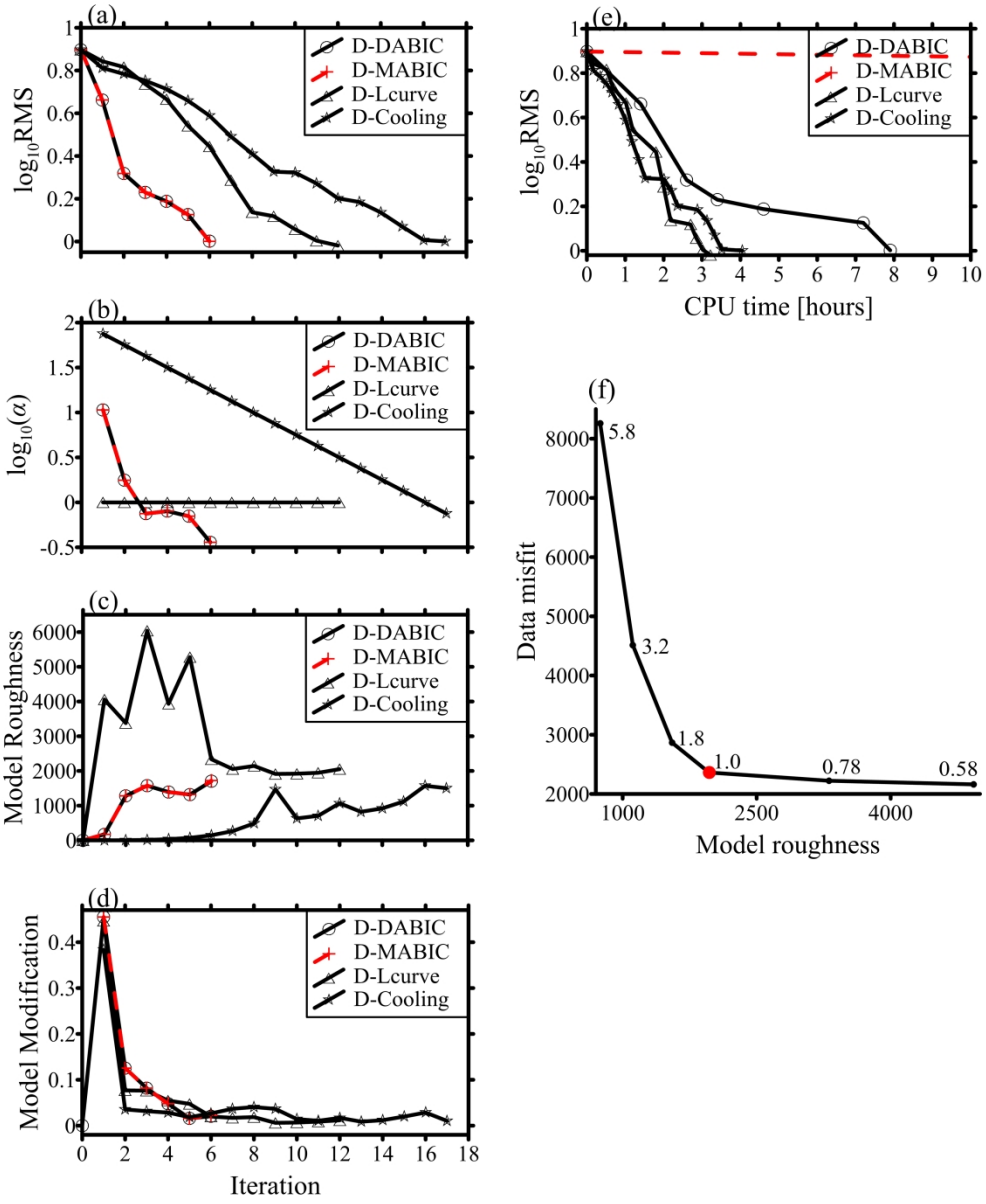


Figure 9. Inversion process metrics for the methods investigated and compared in this study, including (a) RMS misfit on a logarithmic scale ($\log_{10}$RMS), (b) regularization parameter on a logarithmic scale (log10(α)), (c) model roughness, and (d) model modification, each plotted as a function of iteration, as well as (e) RMS misfit on a logarithmic scale ($\log_{10}$RMS) as a function of CPU time. Additionally, (f) presents the L-curve plot used in the D-Lcurve inversion to determine the 'optimal' model shown in Figure 8d.

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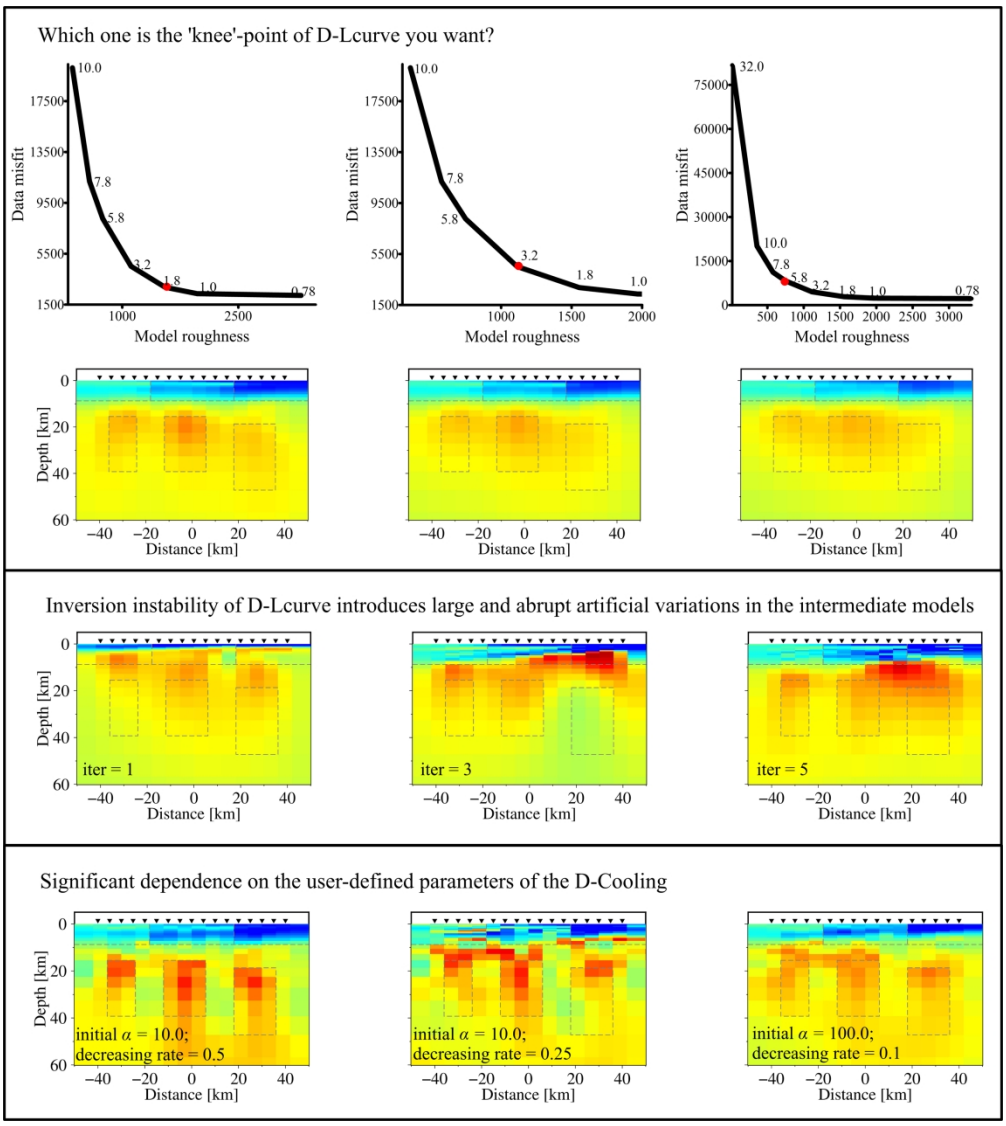


Figure 10. Partial trial-and-error solutions for D-Lcurve and D-Cooling. The top panel illustrates the dependence of the D-Lcurve's "knee" point on the selected interval range. The middle panel demonstrates that the regularization applied in D-Lcurve may not ensure solution stability, leading to abrupt changes that require more iterations for correction. The bottom panel highlights the strong dependence of D-Cooling on manually specified parameters

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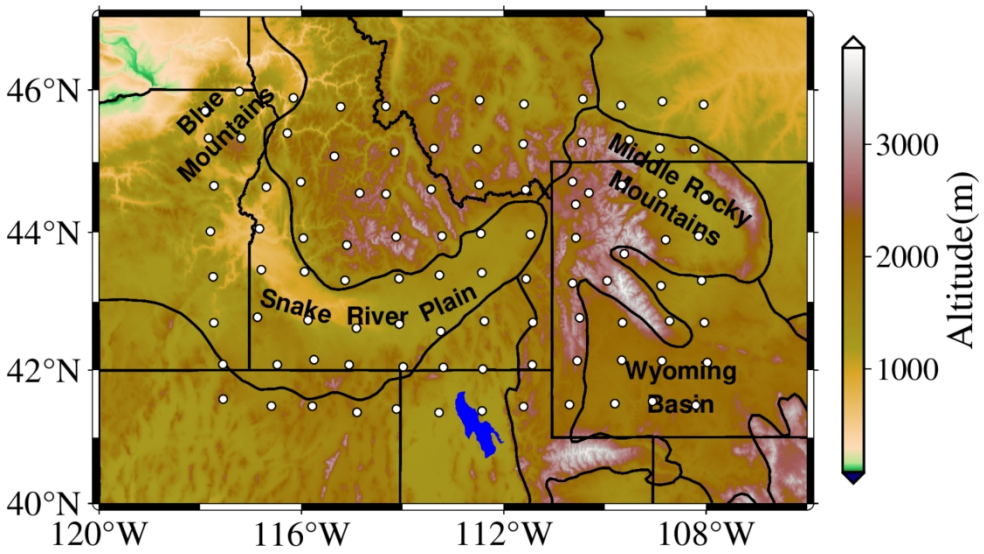


Figure 11. Map of study area, showing topography, MT site locations (white dots), main physiographic provinces, and key elements of the tectonic setting.

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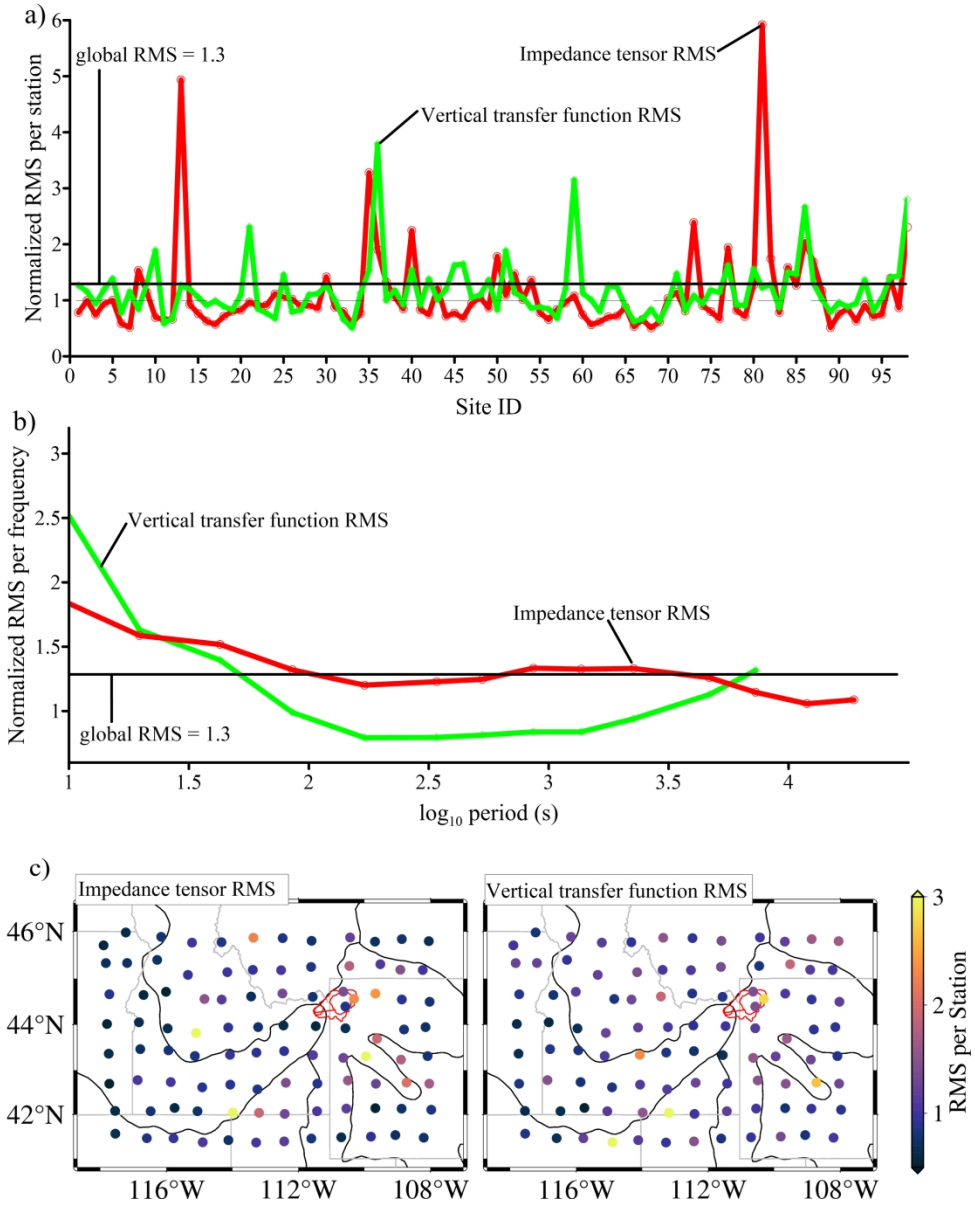


Figure 12. Normalized root-mean-square (RMS) statistics for the response of the final inversion models. (a) RMS by site, (b) RMS by frequency, and (c) RMS by spatial distribution.

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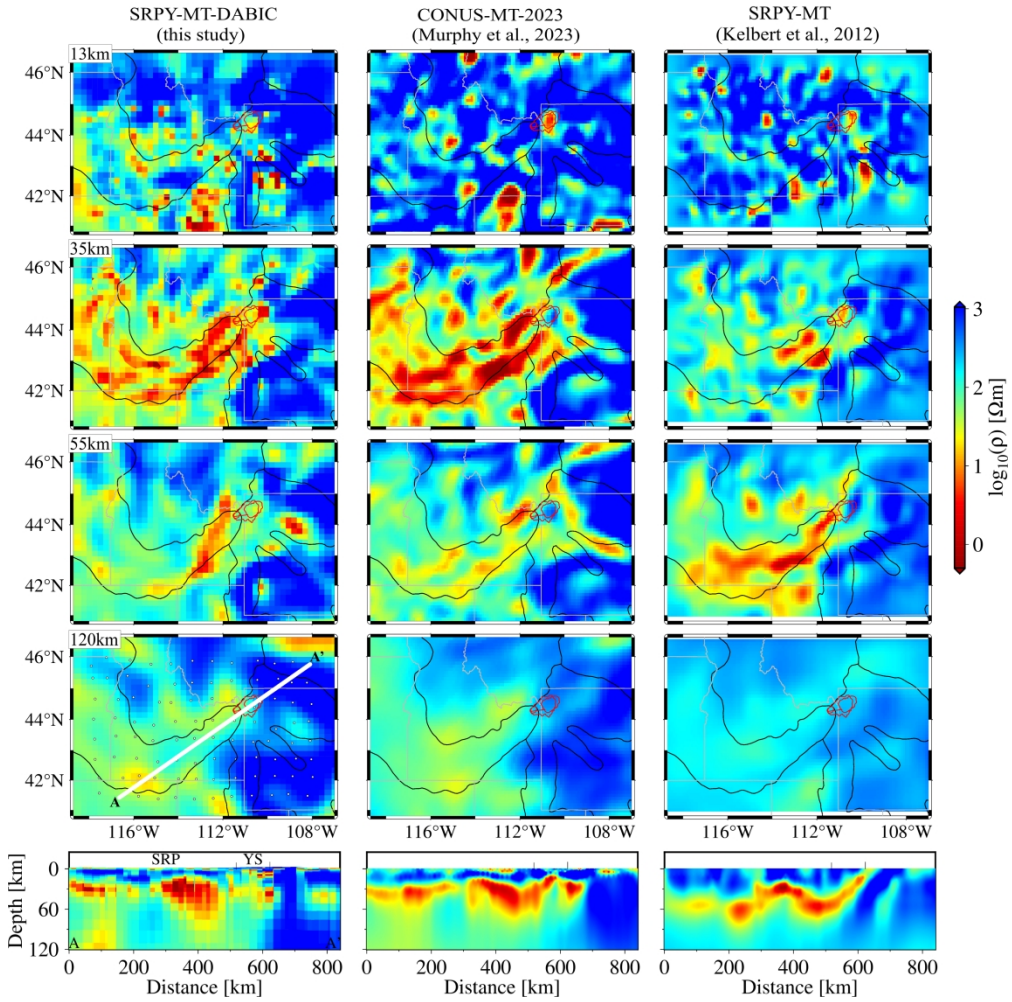


Figure 13. Electrical resistivity structures from three models: the SRPY-MT-DABIC model of this study, CONUS-MT-2023 (Murphy et al., 2023), and SRPY-MT (Kelbert et al., 2012). The top four rows display H-CSs at depths of 13 km, 35 km, 55 km, and 120 km, respectively. In the 120 km-depth H-CS of SRPY-MT-DABIC, white dots denote the locations of MT sites. The bottom row shows V-CSs along the A–A' profile from the above models. In the V-CSs, SRP refers to the Snake River Plain, and YS marks the Yellowstone volcanic region. The two vertical black lines near "YS" indicate the intersections of the A–A' profile with the Yellowstone caldera boundary (outlined in red in the H-CSs). Major physiographic provinces are outlined in white, and key elements of the tectonic settings are outlined in black.

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Table 1. Computation times for different inversions

Method	CPU time per trial (hour)	# of trials required	Total CPU time (hour)
D-DABIC	7.9	1	7.9
D-MABIC	553.6	1	553.6
D-Lcurve	~1 to 4	16	33.8
D-Cooling	~1 to 4	12	26.2

For Peer Review

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DATA AND MATERIALS AVAILABILITY

Data associated with this research are available and can be obtained by contacting the corresponding author.

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